SC- and PC-NEVPT2 and QD-NEVPT2 in free-breakfast

Energies, analytic nuclear gradients and Hessians, and nonadiabatic couplings, density-fitted, on the GPU

free-breakfast implementation notes

7 November 2026

Abstract

This document describes the `src/nevpt2` layer of free-breakfast: the strongly contracted (SC) and partially contracted (PC) n-electron valence state perturbation theory (NEVPT2) energies of a CASSCF reference, their quasidegenerate (QD) effective Hamiltonians over a state-averaged CASSCF, the analytic nuclear gradient of every one of them, the analytic nuclear Hessian of single-state SC and PC, the QD interstate coupling and derivative (nonadiabatic) coupling, and what a QD minimum-energy conical intersection search needs. Everything is density fitted and runs on the device. The organising idea is that every correlation quantity is a function of only three objects, the MO one-electron matrix h, the fitted three-index tensor B and the CI vectors, so one reverse pass through the energy gives its adjoints $(\bar{h}, \Lambda, \bar{y})$, and from there each derivative is the CASSCF machinery the tree already has: one Z-vector solve over the CI-relaxed Hessian and one or two passes of the density-fitted derivative integrals. The Hessian is the same idea one order up: the energy's tangent along each coordinate's first-order response, forward over the reverse pass, against $3N$ CP-CASSCF responses and the gradient's own multipliers. No derivative is written by hand beyond a few scalar finishes, and no finite difference appears in any production path except those that are exact: the Hessian-vector products' bars at fixed multipliers are polynomials of degree at most five along a line, and three symmetric differences recover their derivative with no truncation error.

Contents

1 Scope and layering **1**

2 Notation **2**

3 NEVPT2 and QD-NEVPT2 **3**
- 3.1 The zeroth-order Hamiltonian and the classes . . . 3
- 3.2 The SC energy . . . 3
- 3.3 The PC energy . . . 4
- 3.4 QD-NEVPT2 . . . 4

4 The energy on the device **5**
- 4.1 Density-fitted integrals, slab by slab . . . 5
- 4.2 RDMs: dm3 and the contracted 4-RDMs . . . 5
- 4.3 The SC and PC class walks . . . 6
- 4.4 The QD walks and the amplitude hand-over . . . 7
- 4.5 How the pipeline drives it, and the call graph . . . 8

5 The reverse passes **8**
5.1 One derivative rule . . . 8
5.2 The SC finish . . . 8
5.3 The PC solve, backwards . . . 11
5.4 The QD reverse passes . . . 11

6 The single-state gradient **12**
6.1 The energy as a function of (h, B, C, c) . . . 12
6.2 The adjoints and the orbital source term . . . 12
6.3 The pseudocanonical multipliers . . . 13
6.4 Lagrangian, Z-vector, gate and nuclear term . . . 13

7 The single-state nuclear Hessian **15**
7.1 The $2n + 1$ form . . . 15
7.2 The parameter response and the tangent . . . 15
7.3 The Hessian-vector product . . . 16
7.4 Assembly . . . 17
7.5 Where there is no Hessian: the refusals . . . 18
7.6 The semi-numerical Hessian and the bilinearity check . . . 19
7.7 In the pipeline . . . 19

8 The QD-NEVPT2 gradient **20**
8.1 The functional $\mathcal{E}[W]$ . . . 20
8.2 The Lagrangian on a state average . . . 20
8.3 The W form, and several W at once . . . 21

9 Interstate and derivative couplings **22**
9.1 The interstate coupling . . . 22
9.2 The derivative coupling . . . 22
9.3 The PT2 transition density . . . 23
9.4 Three solves or one: the fold . . . 23

10 Conical intersections **24**

11 Numerical hazards and refusals **25**

12 Verification **26**

13 Map to the code and to the papers **27**

1 Scope and layering

What is implemented.

- SC- and PC-NEVPT2 class energies of one root [1, 2, 3], over a Dyall zeroth-order Hamiltonian [5].
- SC- and PC-QD-NEVPT2 [4]: the effective Hamiltonian of n roots of a state-averaged CASSCF, in the "SA-NEVPT2" scheme of Park [10] (one set of orbitals, core and virtuals canonical over the state-averaged Fock), symmetrised and diagonalised.

- Analytic nuclear gradients of single-state SC and PC-NEVPT2 on a single-state DF-CASSCF [8, 9], and of any SC- or PC-QD-NEVPT2 state on a DF-SA-CASSCF at any weights [10].
- Analytic nuclear Hessians of single-state SC- and PC-NEVPT2 on a single-state DF-CASSCF, all-electron, in Cartesian or caller-given coordinates (§7), with the refusals that keep a non-smooth surface from being reported as a Hessian. Unpublished for any internally contracted MRPT2 as far as we know. There is no QD-NEVPT2 Hessian.
- The QD interstate coupling $\mathbf{h}_{PQ}$, and the full derivative coupling $\mathbf{d}_{PQ} = \langle \Psi_P | \nabla \Psi_Q \rangle$, including the PT2 term [10, 11].
- For conical-intersection searches: $\mathbf{g}_P$, $\mathbf{g}_Q$ and $\mathbf{h}_{PQ}$ at one set of H^{eff} eigenvectors with no refusal of a degenerate pair, and bounds on how far the pair's gap moves when the SA-CASSCF is not fully converged; for PC, the mean gradient alone inside a degenerate CASSCF pair, where the branching plane has no limit.

Layering. `src/nevpt2` sits over `core`, `integrals` (the density-fitted MO transform), `ci` (the strings, sigma vectors and RDMs) and, for derivatives, `response` and `mcscf` (the CASSCF Fock, CI-relaxed Hessian and Z-vector in `response`; the DF gradient, derivative integrals and CASSCF couplings in `mcscf`). It never imports `pipeline` or `hessian`. The nuclear Hessian is assembled one layer up, in `hessian` (`df_sc_nevpt2`, `df_pc_nevpt2`), from this layer's tangents and Hessian-vector products and the CASSCF Hessian's parts. The pipeline's `[nevpt2]`, `[nevpt2_gradient]`, `[nevpt2_nac]`, `[optimize]` and `[hessian]` steps wrap both. Its six sub-modules are:

module	role
`nevpt2.density_fitting`	MO blocks of B, the active $(tu\|vw)$, the ten external blocks slab by slab, and their adjoints onto Λ (§4.1)
`nevpt2.rdm`	dm3 and the integral-contracted 4-RDMs f3ca, f3ac, of one vector or between two, and their adjoints (§4.2)
`nevpt2.energy`	the eight SC and PC class energies, the SC- and PC-QD effective Hamiltonians, the amplitude hand-over, the reverse passes of all four, and the single-state classes' forward tangents and Hessian-vector products (§§4–5, §7.3)
`nevpt2.gradient`	the energy as a function of (h, B, C, c), z^c, the Lagrangians, Z-vectors and nuclear terms; single-state and QD, SC and PC (§§6–8); the builders' forward tangent and the Hessian-vector product in (h, B, c) (§7)
`nevpt2.nac`	the PT2 transition density and the QD derivative coupling (§9)
`nevpt2.intersection`	a QD pair's branching space and gap sensitivities (§10)

Provenance. The energy kernels are a port of the nevpt2-gpu-demo, itself a transcription of PySCF's `mrpt.nevpt2` [16]: every einsum keeps the reference's subscripts, so the two read side by side. The PC energy is validated against block2's `WickICNEVPT2` [17]. The derivative layers are this tree's own. A related device-resident approach, DMRG-QD-SC-NEVPT2 gradients and couplings as one reverse-mode transpose of a contraction graph, is Ref. [18].

2 Notation

Orbitals. C ($n_{\text{ao}} \times n_{\text{mo}}$) maps AOs μ, ν to MOs, ordered core, active, virtual. Core (inactive) labels are i, j, k, l, active t, u, v, w (and p, q, r, s inside active-only tensors), virtual a, b (and r, s as the external labels of a class), general x, y, z. Counts are n_c, n_a, n_v, n_{mo}. A rotation is $C \to Ce^{\kappa}$, κ real antisymmetric.

Integrals. $h = C^{\mathsf{T}} h^{\mathrm{AO}} C$. With three-centre integrals $(\mu\nu|P)$ and the Coulomb metric $(P|Q) = LL^{\mathsf{T}}$,

$$B^L_{xy} = \sum_P (L^{-1})_{LP} \left(C^{\mathsf{T}}(\cdot\,|P)\,C\right)_{xy}, \qquad (xy|zw) = \sum_L B^L_{xy} B^L_{zw}, \tag{1}$$

the density-fitting approximation [15]. Every two-electron quantity below is built from B through such products.

Densities. Spin-summed E_{xy}; active RDMs $\Gamma^{(1)}_{tu} = \langle E_{tu}\rangle$, dm2 $= \langle E_{tu}E_{vw}\rangle$, dm3 $= \langle E_{tu}E_{vw}E_{xy}\rangle$, in PySCF's unreordered layout. Between a bra $\langle M|$ and a ket $|N\rangle$, $\gamma^{(MN)}_{tu} = \langle M|E_{tu}|N\rangle$ and so on, unsymmetrised.

Fock matrices. The inactive Fock and the generalised Fock are

$$F^I_{xy} = h^{\mathrm{eff}}_{xy} = h_{xy} + \sum_j \big[2(xy|jj) - (xj|jy)\big], \qquad f_{xy} = F^I_{xy} + \sum_{tu} \gamma_{tu}\big[(xy|tu) - \tfrac{1}{2}(xt|uy)\big], \tag{2}$$

and the orbital energies are $\varepsilon_x = f_{xx}$ for x core or virtual. On a state average $\gamma \to \gamma^{\mathrm{SA}} = \sum_r w_r \Gamma^{(1)}_r$.

3 NEVPT2 and QD-NEVPT2

3.1 The zeroth-order Hamiltonian and the classes

NEVPT2 partitions $H = H^{(0)} + V$ with Dyall's Hamiltonian [5]

$$H^D = \sum_i \varepsilon_i E_{ii} + \sum_a \varepsilon_a E_{aa} + \sum_{tu} h^{\mathrm{eff}}_{tu} E_{tu} + \tfrac{1}{2} \sum_{tuvw} (tu|vw)(E_{tu}E_{vw} - \delta_{uv}E_{tw}) + C, \tag{3}$$

and splits the perturbation into eight classes $V^{(k)}_l$ by how many electrons the active space gains (k) and from which external orbitals (l) [1]. Their PySCF names are used throughout:

name	space	external tuple	Δ (orbital-energy part of the denominator)
Sr	$S^{(-1)}_r$	a	ε_a
Si	$S^{(+1)}_i$	i	$-\varepsilon_i$
Sijrs	$S^{(0)}_{ijab}$	(i,j,a,b)	$\varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j$
Sijr	$S^{(+1)}_{ija}$	(i,j,a)	$\varepsilon_a - \varepsilon_i - \varepsilon_j$
Srsi	$S^{(-1)}_{abi}$	(a,b,i)	$\varepsilon_a + \varepsilon_b - \varepsilon_i$
Srs	$S^{(-2)}_{ab}$	(a,b)	$\varepsilon_a + \varepsilon_b$
Sij	$S^{(+2)}_{ij}$	(i,j)	$-\varepsilon_i - \varepsilon_j$
Sir	$S^{(0)}_{ia}$	(i,a)	$\varepsilon_a - \varepsilon_i$

Every class but Sijrs has active indices; Sijrs reads no RDM.

3.2 The SC energy

In SC-NEVPT2 each external tuple t has one perturber, $P_t V|\Psi_0\rangle$. Its norm $N_t = \langle\Psi_0|V^\dagger P_t V|\Psi_0\rangle$ and its Dyall element H_t (the active part of $\langle P_t V\Psi_0|H^D - E_0|P_t V\Psi_0\rangle$) are quadratic forms of a slab row of integrals over RDM-built matrices, and

$$E^{\mathrm{SC}} = \sum_k \sum_{t\in k} E_t, \qquad E_t = -\frac{N_t}{\Delta_t + H_t/N_t}, \tag{4}$$

over the perturbers with $|N_t| > 10^{-14}$ (PySCF's `_norm_to_energy` guard; Sijrs and the one-function classes below have none). The matrices are the metrics (dm3, dm2, dm1 and their hole forms) and the Dyall blocks `a16`, `a22`, ... of PySCF, which for Sr and Si read the 4-RDM only through f3ca and f3ac (§4.2).

One-function classes (#949). Srs at two active electrons and Sij at two active holes span one function per perturber: removing both electrons leaves the active vacuum (filling both holes, the closed shell), so $N_t \propto A_t^2$ for one amplitude $A_t = \sum_{pq} W_{pq}\langle \mathrm{vac}|a_q a_p|\Psi\rangle$ and $H_t = \eta N_t$, η that function's Dyall energy, one number per state and class. Every CAS(2,2) has both. PySCF's guard drops such a perturber wherever A_t, a smooth signed function of the geometry, passes through zero, yet its QD couplings $T^t_{IJ} \propto A^I_t A^J_t$ are linear in A_t there: that was a 3–6 $\times 10^{-8}\, E_h$/bohr SC-QD gradient miss at ethylene's Py (#894). So a one-function class (told by tr dm1: Srs at 2, Sij at $2n_a - 2$) keeps every perturber, with $c_t = -1/(\Delta_t + \eta)$ and

$$\eta = \frac{\langle X, Y\rangle}{\langle Y, Y\rangle}, \tag{5}$$

X the class's Dyall block (a7, a9) and Y its metric (rm2, hdm2) in the pairing the H and norm einsums contract. On a state's own RDMs $X = \eta Y$ to 10^{-15}, so η is every perturber's H_t/N_t read with no division by a small N_t. Off that structure (a raw direction in a test) η is the least-squares ratio, one smooth function that every pass differentiates exactly. PC has no guard and is unchanged.

3.3 The PC energy

PC-NEVPT2 keeps the span of each class's internal functions [3]. For a class with metric S, Dyall matrix K, one-body metric S_1 and a coupling factor c, a tuple's slab row x_t and its one-body row h_t give

$$E_t = -c\, b_t^{\mathsf{T}} (K + \Delta_t S)^{+} b_t, \qquad b_t = S x_t + S_1 h_t, \tag{6}$$

$(\cdot)^+$ the inverse on the span of S that the metric cut keeps. S, K and S_1 are the same matrices SC builds (SC contracts each against one row; PC solves with them), so PC is "SC without the final contraction":

class	basis, d	S	K	S_1	c
Sr	(p,q,r), n_a^3	dm3$[r,p,q,b,a,c]$	a16	dm2$[r,p,q,a]$	1
Si	(p,q,r), n_a^3	dm3_h$[r,p,q,b,a,c]$	a22	dm2_h$[r,p,q,a]$	1
Sijr	p, n_a	hdm1	a3	—	$2\,\mathrm{direct}-\mathrm{exchange}$
Srsi	p, n_a	dm1	k27	—	$2\,\mathrm{direct}-\mathrm{exchange}$
Srs	(p,q), n_a^2	rm2$[p,q,b,a]$	a7	—	$\frac{1}{2}$
Sij	(p,q), n_a^2	hdm2	a9	—	$\frac{1}{2}$
Sir	$(p,q)\oplus(p,q)$, $2n_a^2$	spin-coupling blocks	a12, a13	[2 dm1; −dm1]	1
Sijrs	none	—	—	—	SC's

3.4 QD-NEVPT2

Over n roots $|M\rangle$ of a state-averaged CASSCF sharing one set of orbitals, with ε from f^{SA} for every root (Zaitsevskii–Malrieu multipartitioning [6], Park's SA-NEVPT2 scheme [10]), each root has its own perturber space and Dyall Hamiltonian, and the couplings are $C_{MN} = \langle M|V|\Psi^{(1)}_N\rangle$. With $\Psi^{(1)}_N$

the ket's first-order function:

$$\text{SC:}\quad C_{MN} = -\sum_t \frac{\bar{T}^t_{MN}}{\Delta_t + H^N_t/N^N_t}, \qquad \bar{T}^t_{MN} = \tfrac{1}{2}(T^t_{MN} + T^t_{NM}), \tag{7}$$

$$\text{PC:}\quad C_{MN} = -c\sum_t b^{M\leftarrow N\,\mathsf{T}}_t (K_N + \Delta_t S_N)^{+} b^N_t, \qquad b^{M\leftarrow N}_t = S^{(N,M)} x_t + S_1^{(N,M)} h_t. \tag{8}$$

T^t_{MN} is the class's norm einsum over pair (M, N)'s transition RDMs; the terms of it that read no RDM carry $\langle M|N\rangle = 0$ and are dropped. The mean over (M, N) and (N, M) in SC undoes PySCF's factor 2 on the Hermitian cross terms of Sr, Si and Sir, which holds only for a state's own RDMs. $S^{(N,M)}$, $S_1^{(N,M)}$ are the PC metrics over pair (N, M)'s transition RDMs (bra N, ket M); S_1 must be that orientation. PC has no mean, so its C is not symmetric. Sijrs is diagonal. The effective Hamiltonian is symmetrised as in MS-CASPT2 [10]:

$$H^{\text{eff}}_{MN} = E_M\delta_{MN} + \tfrac{1}{2}(C_{MN} + C_{NM}), \qquad H^{\text{eff}} R_P = E_P R_P, \quad R^{\mathsf{T}} R = 1, \tag{9}$$

each column of R signed so its largest-magnitude component is positive.

4 The energy on the device

4.1 Density-fitted integrals, slab by slab

`mo_blocks` forms the five MO blocks B_{aa}, B_{ca}, B_{va}, B_{cv} and B_{vc} (B_{cv} with its pair swapped, so every batched index is the first of its pair). No $(Q|vv)$ block exists: no class has two virtuals in one charge distribution. The ten external blocks are built a slab at a time along their batch axis, so no whole external block is ever on the device:

$$\mathcal{C}_{xyzw} = (xy|zw) = \sum_Q X_{Q,xy}\, Y_{Q,zw}, \qquad \text{slab} = \text{permute}(\mathcal{C}), \tag{10}$$

one strided-batched GEMM over y ($\mathcal{C}_y = X_y^{\mathsf{T}} Y$) and one permute kernel per slab; Sijrs needs no permute.

block	physicist slab	X	Y	axes	batch
Sr	$[r,a,b,c] = (rb\|ac)$	B_{va}	B_{aa}	0213	0
Si	$[t,u,i,v] = (it\|uv)$	B_{ca}	B_{aa}	1203	2
Sijrs	$[i,a,j,b] = (ia\|jb)$	B_{cv}	B_{cv}	0123	0
Sijr	$[r,p,j,i] = (rj\|ip)$	B_{vc}	B_{ca}	0312	0
Srsi	$[x,y,i,t] = (xi\|yt)$	B_{vc}	B_{va}	0213	0
SrsiT	$[y,x,i,t] = (xt\|yi)$	B_{va}	B_{vc}	2031	1
Srs	$[r,s,t,u] = (rt\|su)$	B_{va}	B_{va}	0213	0
Sij	$[t,u,i,j] = (it\|ju)$	B_{ca}	B_{ca}	1302	2
Sir1	$[r,t,i,u] = (ri\|tu)$	B_{vc}	B_{aa}	0213	0
Sir2	$[r,u,t,i] = (rt\|iu)$	B_{va}	B_{ca}	0312	0

Adjoints. With $\bar{\mathcal{C}}$ the slab bar permuted back, $\bar{X}_y \mathrel{+}= Y\bar{\mathcal{C}}_y^{\mathsf{T}}$ and $\bar{Y}_w \mathrel{+}= X_b\bar{\mathcal{C}}_w$, two strided-batched GEMMs written straight into $\Lambda = \partial E/\partial B$ ($n_{\text{aux}} \times n_{\text{mo}}^2$, B's own layout) at each block's corner, so no block bar is ever formed. A slab's bar is consumed by its call and freed before the next slab is built.

4.2 RDMs: dm3 and the contracted 4-RDMs

The 4-RDM never exists. Sr and Si read it only contracted with $(tu|vw)$:

$$\mathrm{f3ca}_{pqrstu} = \sum_{ijk} (ki|uj)\,\langle E_{pq}E_{rs}E_{tj}E_{ki}\rangle, \qquad \mathrm{f3ac}_{pqrstu} = \sum_{ijk} (ik|jt)\,\langle E_{pq}E_{rs}E_{ju}E_{ik}\rangle, \tag{11}$$

(here i, j, k active). `nevpt2::rdms` builds dm3, f3ca and f3ac (n_a^6 each), tiled over determinants K against a byte budget. Per tile, one thread per determinant gathers

$$R_{tu,K} = \langle K|E_{tu}|c\rangle, \qquad L2_{pqrs,K} = \langle c|E_{pq}E_{rs}|K\rangle \tag{12}$$

from the strings' own replacement lists (Knowles–Handy [14]), then dm3 $+= L2\,R^{\mathsf{T}}$ is one GEMM; per order the consume $W = (tu|vw)\cdot L2$ is n_a strided-batched GEMMs and $f3 += L2\,W^{\mathsf{T}}$ one GEMM. After the last tile a contraction of $(tu|vw)$ with dm3 (fdm2) and PySCF's wedge reconstruction (the commutator $E_{pq}E_{rs} = E_{rs}E_{pq} + \delta_{qr}E_{ps} - \delta_{ps}E_{rq}$) run once per order. Nothing is atomic, so the result does not depend on the run. The cost is $O(N_{\mathrm{det}}n_a^8)$ in GEMMs, the same order as forming the 4-RDM, with $O(N_{\mathrm{det}}n_a^4)$ memory per tile.

Between two vectors. The bra/ket build gathers $L2$ over the bra and R over the ket, so $L2\,R^{\mathsf{T}} = \langle M|EEE|N\rangle$; dm1 and dm2 come from one GEMV each per tile. The consume reads the ket's own $L2$ (a second gather), and the ca order's fdm2 reads the transition element that hermiticity let the one-vector build skip. With `bra == ket` it is the one-vector build operation for operation. QD reads f3 on the diagonal only, so the pair loop passes null f3 pointers and only dm1–dm3 are built, about a third of the GEMM work.

Adjoints. `rdms_adjoint` runs the build backwards over the same tiles: the wedge and fdm2 first, then per tile each digest acc $+= L2\,B^{\mathsf{T}}$ as two GEMMs ($\bar{L2} += \overline{\mathrm{acc}}\,B$, $\bar{B} = \overline{\mathrm{acc}}^{\mathsf{T}}L2$), each consume as two loops of batched GEMMs, and one gather onto the CI bar over each determinant's replacement lists. dm1 and dm2 have no bars of their own: since $\sum_t E_{tt} = N$ on the ket,

$$\mathrm{dm2}_{pqrs} = \tfrac{1}{N}\sum_t \mathrm{dm3}_{pqrstt}, \qquad \mathrm{dm1}_{pq} = \tfrac{1}{N^2}\sum_{rt} \mathrm{dm3}_{pqrrtt}, \tag{13}$$

so their bars fold into $\overline{\mathrm{dm3}}$ along those traces, and one adjoint carries every RDM. Neither a sigma vector nor any symmetry of a bar is assumed.

4.3 The SC and PC class walks

Each class builds its RDM-only intermediates once, then walks its external index in batches: the slab is built, contracted by the class's einsums (`recipes::einsum`, with the reference's subscripts unchanged) into per-perturber N_t and H_t, read back and finished on the host, Eq. (4). Srsi pairs (r, s) with (s, r) across slabs, so it fills its whole (v, v, c) tables first.

PC solves each class once, since S, K and S_1 do not depend on the tuple (Algorithm 1). Each slab is then one einsum into the eigenbasis.

S is singular by construction (Sr and Si drop 44 of 64 modes at CAS(4,4)). The cut sits at least 13 times above every mode round-off or the RDMs decide, measured; above it the real modes may run through the cut with no gap, and carry no energy there (PSB3, HF with its F 1s active: the energy moves by under $3 \times 10^{-9}\,E_h$ with the cut anywhere in $10^{-15} \ldots 10^{-12}$; `src/nevpt2/energy/README.md`, #853), so there is no gap test. A refused class (no positive σ, or a non-positive denominator) fails the call (naming the class, and for QD the state); it is never summed in as zero.

Algorithm 1 A PC class (`solve_class`, then the walk)

Input: S, K, S_1 ($d \times d$, $d \times n_1$) of one state; the class's slabs

1: $S \leftarrow \operatorname{sym} S$, $K \leftarrow \operatorname{sym} K$ ▷ K is symmetric only to the CI's convergence
2: $S = V\sigma V^{\mathsf{T}}$; refuse if $\sigma_{\max} \le 0$; keep $\sigma_k > \max(10^{-13}, 100\,\varepsilon_{\mathrm{mach}}\sqrt{d}, f)\,\sigma_{\max}$ ▷ no gap test (#853); f the floor
3: $X \leftarrow V_{\mathrm{kept}}\sigma_{\mathrm{kept}}^{-1/2}$; $K' = X^{\mathsf{T}}KX = W\lambda W^{\mathsf{T}}$; $U \leftarrow XW$
4: $\mathcal{T} \leftarrow SU$; $t_1 \leftarrow S_1^{\mathsf{T}}U$
5: **for** each slab, each tuple t **do**
6: $\quad y_t \leftarrow \mathcal{T}^{\mathsf{T}}x_t + t_1^{\mathsf{T}}h_t$ ▷ one einsum per slab
7: $\quad$ refuse if $\min_k(\lambda_k + \Delta_t) \le 0$
8: $\quad E_t \leftarrow -c\sum_k y_{tk}^2/(\lambda_k + \Delta_t)$

The conditioning floor (#917, #923, #1020). f is `EnergyInputs::pc_floor`, relative to each metric's $\sigma_{\max}$, and each class reports `cut` (the largest dropped σ) and `kept` (the smallest kept), over $\sigma_{\max}$. The library default is $f = 0$, the round-off cut alone, so every energy, gradient and golden is bitwise what it was before the floor existed. The floor matters at second order, not first. At PSB3, DF-CASSCF(6,6)/cc-pVDZ, real modes sit at about 5×10^{-13} of $\sigma_{\max}$. Along the Hessian's physical directions the first-order bars repeat to 10^{-10} of their largest component, but a central-difference ladder of the PC adjoint never converges: near-cut kept modes swing λ through $-\Delta$ as $1/\sigma$. At $f = 10^{-10}$ the ladder is clean at FC and at the 90° twist. The energy moves by 1.2×10^{-10} (FC) and $5.8 \times 10^{-9}\,E_h$ (twist), and about 1.7×10^{-7} at HF CAS(10,7). A library-wide 10^{-10} would move the PC-QD goldens and NACs past their bars, so it is applied only where needed: the pipeline's PC `[hessian]` runs its whole chain at `pc_floor` $= 10^{-10}$ by default. The QD entry points take a trailing `pc_floor = 0` and set it on every state's inputs; nothing in the pipeline raises it there. One field read by the one `solve_class` means the energy, its reverse pass and its tangent always differentiate the same function.

4.4 The QD walks and the amplitude hand-over

`sc_effective_hamiltonian` and `pc_effective_hamiltonian` take $n \times n$ pair inputs, entry (M, N) carrying the shared integrals and pair (M, N)'s transition dm1–dm3 (f3 on the diagonal only). Each class walks its slabs once for every pair. For SC, per slab, one table per given pair accumulates T^t_{MN} and one per given state N^N_t, H^N_t; the Dyall blocks are built from each state's own RDMs only. For PC, each ket N's $(S_N, K_N, S_{1,N})$ is solved once, the transitions' $\mathcal{T}_{M\leftarrow N} = S^{(N,M)\mathsf{T}}U_N$ and $t_{1,M\leftarrow N} = S_1^{(N,M)\mathsf{T}}U_N$ are built once per pair, and

$$C^t_{MN} = -c\sum_k \frac{y_{M\leftarrow N,tk}\; y_{N,tk}}{\lambda_{N,k} + \Delta_t}, \qquad y_{M\leftarrow N,t} = \mathcal{T}^{\mathsf{T}}_{M\leftarrow N}x_t + t^{\mathsf{T}}_{1,M\leftarrow N}h_t, \tag{14}$$

with Sijr and Srsi's "2 direct − exchange" as the bilinear $\sum_t (2y_{M\leftarrow N,t}y_{N,t} - y_{M\leftarrow N,t}y_{N,t'})/(\lambda + \Delta_t)$.

The hand-over. Both calls can return each state's first-order amplitudes: for SC $c_t = -1/(\Delta_t + H^N_t/N^N_t)$ (zero where the guard drops the perturber; η of Eq. (5) in place of H/N for a one-function class, which drops none), so $\Psi^{(1)}_N \ni c_t P_t V|N\rangle$; for PC U_N and $g_{tk} = y_{N,tk}/(\lambda_{N,k} + \Delta_t)$, so $\Psi^{(1)}_N \ni -\sum_k g_{tk}\sum_\mu U_N[\mu, k]\phi^N_\mu$. A later call with its diagonal omitted reads them instead of walking the states again: n states take one diagonal call, then one call per pair over (M, N) and (N, M) alone, holding two transition sets at a time and never n^2. The PT2 transition density (§9.3) contracts the same amplitudes.

4.5 How the pipeline drives it, and the call graph

`pipeline/nevpt2` (`[nevpt2]`) runs only over DF integrals and a closed-shell singlet, and refuses integral-direct ERIs and open shells before anything is queued.

Per root (state-specific, SC and PC from the same inputs): γ_r, Γ_r of the root's CI vector alone; the root's own Fock $F = F^I + F^A(\gamma_r)$, with the core and virtuals canonicalised over it ($C_r = CR$, R block-diagonal), so ε are F's eigenvalues and *not* the CASSCF's state-averaged `mo_energies` (the active orbitals are not rotated to natural orbitals, as PySCF does, since both variants are invariant to that); $R^\mathsf{T} F^I R$ sliced into h and the three `h1e_v` blocks; the MO blocks of B over C_r; dm1 $= \gamma_r$, dm2 $= \Gamma_r + \delta_{qr}\gamma_{ps}$, and dm3, f3ca, f3ac by `nevpt2::rdms`; then `class_energies` and `pc_class_energies` over the same inputs. A PC refusal is logged and that root keeps its SC.

With `quasi_degenerate`: one shared set of orbitals, the CASSCF's own, core and virtuals canonical over the state-averaged Fock, its `mo_energies` as ε. So *the QD diagonal differs from the per-root results*: it is root r's NEVPT2 on the shared orbitals, not on its own. Each root's own RDMs are kept from the per-root pass (they do not depend on the core–virtual rotation). Then the diagonal-only call, the per-pair calls over the transition dm1–dm3 with null f3, Eq. (9), and its eigenvectors, each signed so its largest component is positive. A PC-QD refusal leaves PC-QD empty and SC-QD standing.

Figure 1 is `Nevpt2Step::run` as a DAG, from the (SA-)CASSCF's output to the four energies; Figure 2 is what the four entry points call inside the energy layer. Blue nodes are the layer's entry points, orange the RDM builds that dominate as the active space grows. With `quasi_degenerate` the per-root pass still runs first: the QD diagonal reads its RDMs, though not its orbitals.

5 The reverse passes

5.1 One derivative rule

`einsum_adjoint` takes a forward einsum's subscripts, operands, output shape and scale unchanged, and writes operand k's bar as the same einsum with the output and operand k swapping roles. An operand that repeats a label takes its bar over its distinct labels and is scattered onto the diagonal; a strided view takes its bar packed and one GEMM-add places it. Every class lists its norm and H terms once, as data, and both directions read the list, so no derivative of a class is transcribed by hand. The NumPy slice assignments of the reference (`a16[:, i, :, :, :, i] += fdm2` and two in a22) have the gather fdm2$^\mathsf{T}$-bar = `einsum("pirabi->prab", a16_bar)` as their adjoint, since an input label may repeat.

5.2 The SC finish

The only hand-differentiated expression on the SC side is Eq. (4)'s finish. With $T_t = -1/(\Delta_t + H_t/N_t)$ [8]:

$$P_t = \frac{\partial E}{\partial N_t} = 2T_t + \Delta_t T_t^2, \qquad Q_t = \frac{\partial E}{\partial H_t} = T_t^2, \qquad R_t = \frac{\partial E}{\partial \Delta_t} = N_t T_t^2, \tag{15}$$

zero for a dropped perturber. A one-function class puts $P_t = c_t$ on N_t, nothing on H_t, $\bar{D}_t = \bar{H}_{JJ} N_t c_t^2$ (plus the QD share) on Δ_t, and $\bar{\eta} = \sum_t \bar{D}_t$ onto its blocks after the walk: $\bar{\eta}\, Y/\langle Y, Y\rangle$ on X and $\bar{\eta}\,(X - 2\eta Y)/\langle Y, Y\rangle$ on Y. P and Q go back through the norm and H einsums onto the slabs (and so Λ), the one-body rows and the blocks' bars $M_G = \partial E/\partial G$; R goes onto the orbital energies Δ_t reads. Sijrs has its own kernel. Srsi walks twice, since its finish pairs entries across slabs.

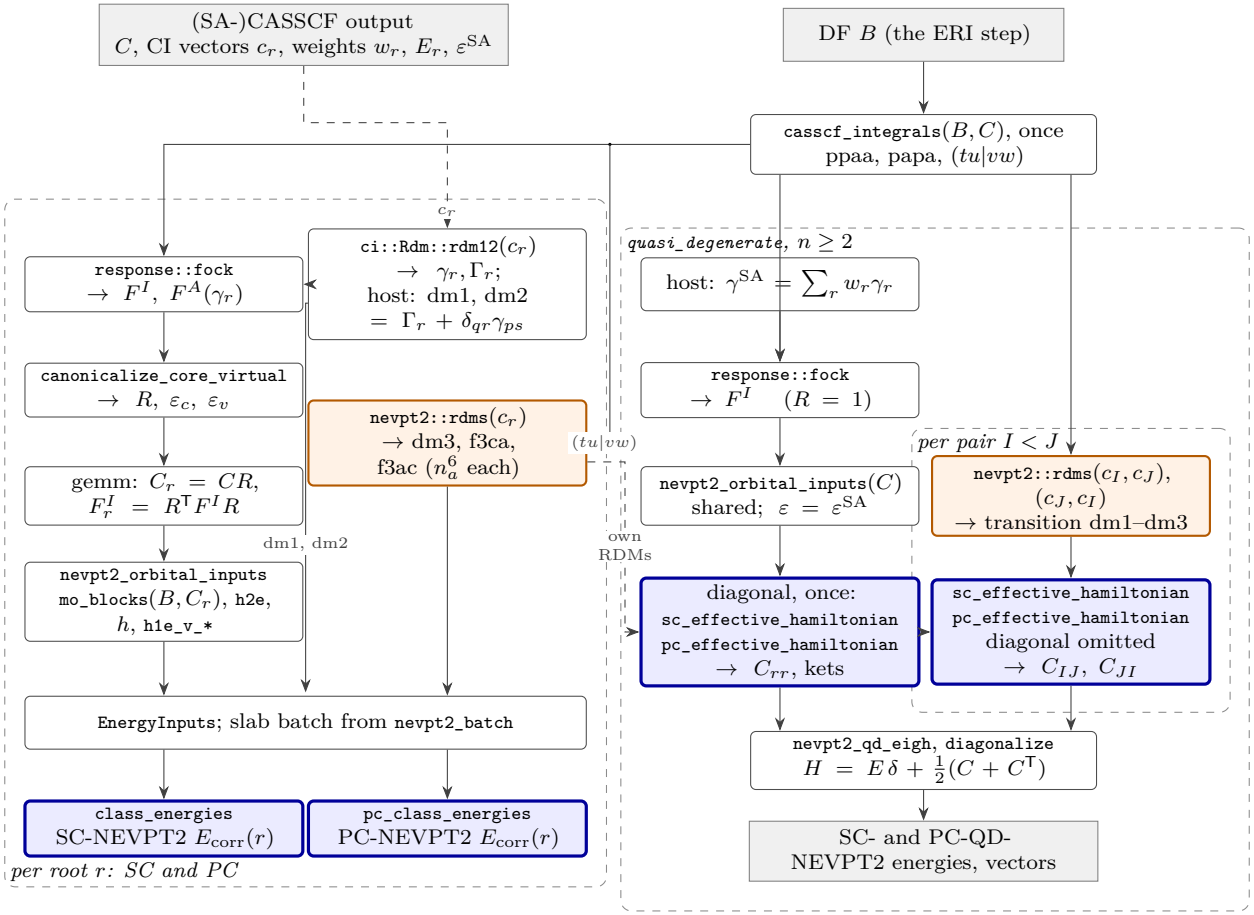


Figure 1: `Nevpt2Step::run` (`src/pipeline/nevpt2/nevpt2.cpp`). Left, every root's own orbitals and RDMs, for SC- and PC-NEVPT2. Right, the shared state-averaged orbitals for QD: the diagonal is walked once and hands its first-order wave functions (the kets) and its inputs to the pair calls, which build only transition RDMs and walk only the transition norms. C and the CI vectors reach every node that names them; dashed edges carry data from the CASSCF or across the two passes.

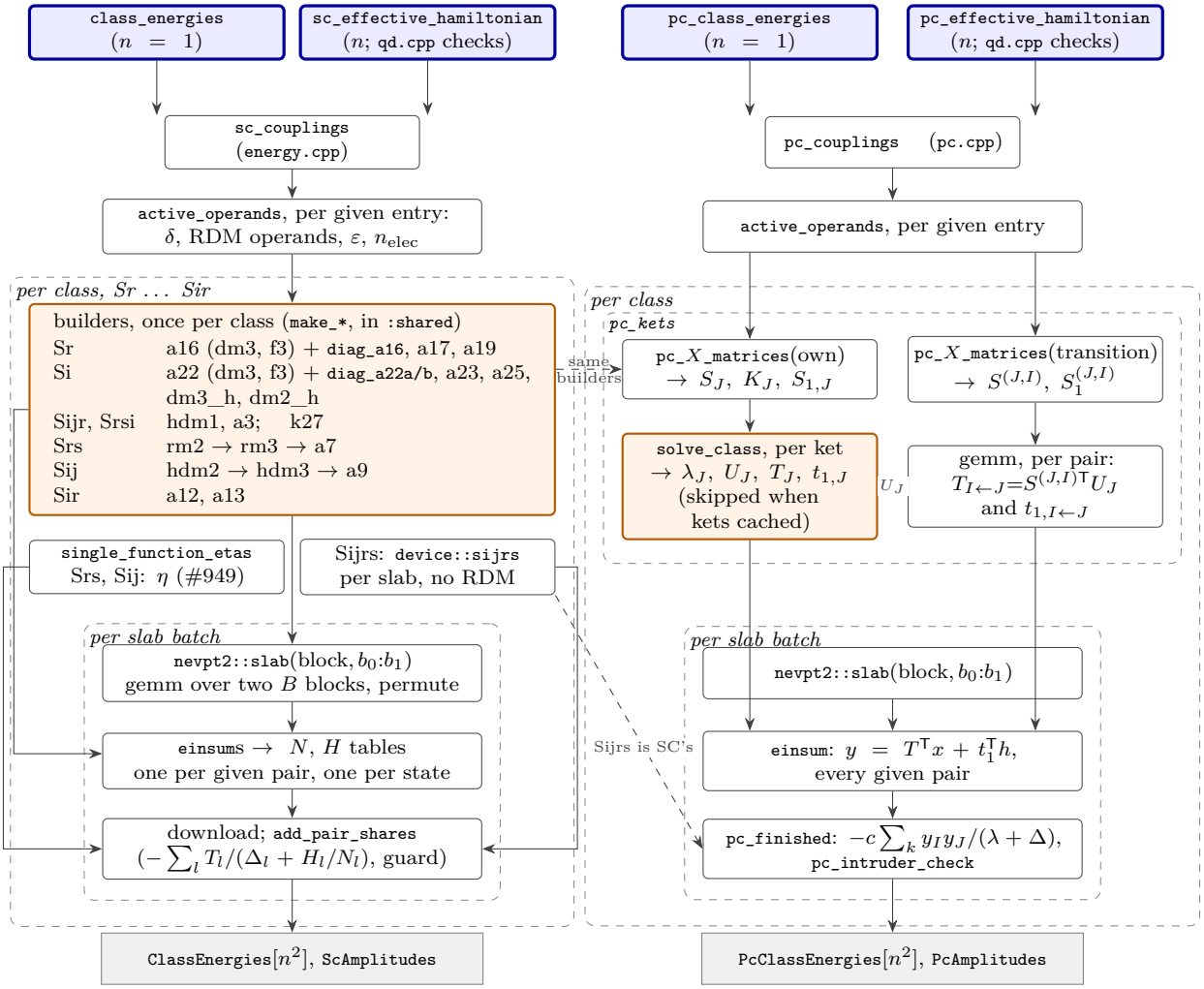


Figure 2: Inside `free_breakfast.nevpt2.energy`. Each single-state call is its effective Hamiltonian's walk at $n = 1$. A class builds its RDM intermediates once, then walks its external index in batches, building one slab of the density-fitted integrals at a time. PC solves each ket's metric once per class (`solve_class`) and reads the slab only through one einsum into that eigenbasis. With the diagonal omitted, both read the kets from the amplitudes a diagonal call returned and solve nothing.

The sweep. The block bars are swept back through the builders that formed them, in reverse topological order (an intermediate's bar is complete before it is swept), onto $\bar{h}$, the active $\overline{(tu|vw)}$ and $\overline{\mathrm{dm1}}, \ldots, \overline{\mathrm{f3ac}}$. It has two halves: the Dyall blocks, which read h and $(tu|vw)$ and only a state's own walk has, and the metrics, which read only RDMs and are all a QD transition takes. Intermediates are recomputed, never kept.

Identities from the bars. Each class energy is homogeneous of degree one in (N, H), so

$$\sum_t (P_t N_t + Q_t H_t) = \sum_G \langle M_G, G\rangle = E_k \tag{16}$$

(Park's Eq. 40 [8]); and at fixed orbital energies E is homogeneous of degree 4 in $(B, h_v^{1/2})$, so $\langle \Lambda, B\rangle + 2\sum\langle \bar{h}_v, h_v\rangle = 4E$, h_v the one-body rows. Both are step-free checks of the bars.

5.3 The PC solve, backwards

With $A_t = K + \Delta_t S$ and $x = A_t^+ b_t$ the solve the energy ran,

$$\dot{E}_t = -c(2\, x^{\mathsf{T}} \dot{b}_t - x^{\mathsf{T}} \dot{A}_t\, x), \tag{17}$$

so no eigenvalue or eigenvector derivative appears. This holds where S's kept rank is locally constant and K annihilates S's dropped modes, which holds at an eigenstate for its null modes. With real modes dropped at the cut (#853) it is validated at PSB3 against re-converged central differences to max(2× the Richardson bar, $3\sigma_E/h_{\min}$), and keeping those modes does not move it (`src/nevpt2/energy/README.md`). In the eigenbasis, with $g_t = y_t/(\lambda + \Delta_t)$, each tuple gives $\bar{y}_t = -2c\, g_t$, a weight $w_t = c\, g_t$ and $\partial E/\partial \Delta_t = c\, g_t \cdot g_t$; accumulating $M = \sum_t g_t w_t^{\mathsf{T}}$ and $M_\Delta = \sum_t \Delta_t g_t w_t^{\mathsf{T}}$ over the walk,

$$\frac{\partial E}{\partial K} = UMU^{\mathsf{T}}, \qquad \frac{\partial E}{\partial S} = U\bar{\mathcal{T}}^{\mathsf{T}} + UM_\Delta U^{\mathsf{T}}, \qquad \frac{\partial E}{\partial S_1} = U\bar{t}_1^{\mathsf{T}}, \tag{18}$$

symmetrised as the solve symmetrised. Those go back through the einsums that built S, K, S_1 onto SC's block bars, and SC's sweep takes them unchanged. At fixed slabs and orbital energies a PC class is homogeneous of degree one in (S, S_1, K), which gives PC's analogue of Eq. (16).

5.4 The QD reverse passes

Both QD passes differentiate $L = \sum_{MN} \bar{H}_{MN} H_{MN}$ for a given $\bar{H}$; at $n = 1$, $\bar{H} = 1$ they are the single-state passes einsum for einsum.

SC-QD. With $D_N = \Delta + H^N/N^N$, the coupling $-\bar{T}_{MN}/D_N$ has three bars:

$$\frac{\partial L}{\partial \bar{T}_{MN}} = -\frac{\bar{H}_{MN}}{D_N}, \qquad A_N = \sum_M \frac{\bar{H}_{MN}\bar{T}_{MN}}{D_N^2}, \qquad (\bar{N}^N, \bar{H}^N, \bar{\Delta}) \mathrel{+}= \Big(-\frac{A_N H^N}{(N^N)^2},\ \frac{A_N}{N^N},\ A_N\Big), \tag{19}$$

the first split half and half onto the norm einsums of (M, N) and (N, M). The walk runs in three parts: each state's tables, evaluated only; each unordered pair, its norm einsums evaluated and differentiated at once, its metric bars swept onto its transition dm1–dm3 and handed to a callback, then freed; each state's own walk with A_N added. Memory is one pair's transition set at a time, never n^2 sets.

PC-QD. The coupling is bilinear in the two rows, so with $g_M = y_{M\leftarrow N}/(\lambda_N + \Delta)$ and $g = y_N/(\lambda_N + \Delta)$ the bars are $-c\bar{H}_{MN}g$ on $y_{M\leftarrow N}$, $-c\bar{H}_{MN}g_M$ on y_N, $c\bar{H}_{MN}g_M g^{\mathsf{T}}$ on M_N (and Δ times it on $M_{\Delta,N}$), and $c\bar{H}_{MN}g_M \cdot g$ on Δ. Still no eigenvector derivative: $b^{M\leftarrow N}$ lies in S_N's kept range, since a mode annihilating $|N\rangle$ annihilates $\langle N|S^{(N,M)}$. Then $\partial L/\partial S^{(N,M)} = \bar{\mathcal{T}}_{M\leftarrow N}U_N^{\mathsf{T}}$ and $\partial L/\partial S_1^{(N,M)} = U_N \bar{t}_{1,M\leftarrow N}^{\mathsf{T}}$, unsymmetrised, onto pair (N, M)'s transition RDMs. Each class walks its slabs once with every pair, as the forward does.

6 The single-state gradient

6.1 The energy as a function of (h, B, C, c)

`sc_energy` and `pc_energy` rebuild everything the class walk reads from (h, B) in the MO basis of C and the CI vector c:

$$h1e = F^I_{tu}, \quad h1e_v_si = F^I_{ti}, \quad h1e_v_sir = F^I_{ai}, \quad h1e_v_sr = F^I_{at} - \sum_u (au|ut), \quad \varepsilon_x = f_{xx}, \tag{20}$$

dm3, f3ca, f3ac by `rdms`, and dm1, dm2 as dm3's traces (Eq. (13)). Two choices matter:

- ε **moves with** γ**.** It is rebuilt from (h, B, γ) at every point, never an input. Holding it fixed differentiates a different functional, and a check whose two sides share the omission cannot see it.
- **The orbitals are taken as they are.** ε is f's diagonal; off canonical orbitals nothing re-canonicalises. SC-NEVPT2 is not invariant to core–core or virtual–virtual rotations, so the pseudocanonical conditions $f_{ij} = f_{ab} = 0$ belong to the Lagrangian, not the functional.

Beyond the five MO blocks the builders read B_{cc} whole and B_{vv}'s diagonal.

6.2 The adjoints and the orbital source term

(h, B) in the MO basis is the interface because every NEVPT2 input is built from it: the orbital energies, the inactive Fock blocks, the active $(tu|vw)$ and every external slab. So one reverse pass reaches the whole energy, and both the orbital rotations and the nuclear derivative integrals act on (h, B) alone. One reverse pass (§5, then the builders backwards, `rdms_adjoint` and the DF adjoints) gives

$$\bar{h} = \frac{\partial E_{\rm corr}}{\partial h}, \qquad \Lambda^L = \frac{\partial E_{\rm corr}}{\partial B^L}, \qquad y = \frac{\partial E_{\rm corr}}{\partial c}\Big|_{h,B}, \tag{21}$$

$\bar{h}$ and each Λ^L symmetrised last. Along $C \to C(1 + \theta)$, any $\Phi(h, B)$ with symmetric adjoints moves by $\sum \theta_{xy}X_{xy}$,

$$X = 2\,h\,\bar{h} + 2\sum_L B^L\Lambda^L, \qquad A = X - X^{\mathsf{T}} = \frac{\partial \Phi}{\partial \kappa}, \tag{22}$$

X needs B^L_{xq} with x general, which is the full-range B or, without forming it, an AO half-transform:

$$X = 2\,C^{\mathsf{T}}h^{\rm AO}C\,\bar{h} + 2\sum_L C^{\mathsf{T}}J^L(C\,\Lambda^L), \qquad J^L_{\mu\nu} = \sum_P (L^{-1})_{LP}\,(\mu\nu|P), \tag{23}$$

evaluated as $\tilde{\Lambda}_q = \Lambda_q C^{\mathsf{T}}$ per q (one batched GEMM), then $D = B^{\mathsf{T}}_{\rm AO}\tilde{\Lambda}$, the DF-exchange shape, rather than pair by pair. It is exact by construction: $\langle \kappa, A\rangle$ equals the forward tangent $\dot{E}(\kappa)$, a check rather than an assembly route. Every block of A has a reader: the mixed blocks the Z-vector, core–core and virtual–virtual z^c, active–active the gauge identity. A is Park's orbital source term Y' [8].

6.3 The pseudocanonical multipliers

Along $\kappa_{pq} = -\kappa_{qp} = \theta$ within the core or the virtuals, f_{pq} and f_{qp} move by $\theta(\varepsilon_p - \varepsilon_q)$ and nothing else the constraint reads moves at first order. So stationarity of $E_{\mathrm{corr}} + \langle z^c, f\rangle$ (full Frobenius sum, symmetric z^c) decouples pair by pair:

$$z^c_{pq} = z^c_{qp} = -\frac{A_{pq}}{2(\varepsilon_p - \varepsilon_q)}, \qquad p \neq q \text{ both core or both virtual,} \tag{24}$$

Park's Eqs. 56–57 [8] and Eqs. 25–26 [10] in this tree's sign of κ. `zc_multipliers` takes only A and ε, so it serves SC, PC and QD alike.

Degenerate pairs. For $|\varepsilon_p - \varepsilon_q| < 10^{-6}\,E_h$, a pair with $|A_{pq}| > 10^{-8}$ is refused, naming both orbitals; one with A_{pq} under its rounding bound (n^2_{mo} unit roundoffs of $\max|A|$; partners in different irreps, say) has $z^c_{pq} = 0$; any other keeps Eq. (24). PC's A vanishes with the gap, $A_{pq} = (\varepsilon_p - \varepsilon_q)K_{pq}$ with K_{pq} finite, so PC's z^c has the limit $-K_{pq}/2$ where SC's grows as the inverse gap and is refused: this is Park's reason for preferring PC [8]. Dropping PC's z^c there would be a step of $O(z)$ in the gradient, not $O(\mathrm{gap})$ (#974; `src/nevpt2/gradient/README.md`, "Degenerate pairs").

The term z^c adds. $\Phi = \langle z^c, f\rangle$ with $D = 2$ on the core diagonal and γ (or γ^{SA}) on the active block, $g_L = \sum B^L_{rs} D_{rs}$ and $w_L = \sum B^L_{pq} z_{pq}$:

$$\Phi = \langle z, h\rangle + \sum_L w_L g_L - \tfrac{1}{2}\sum_L \mathrm{tr}(zB^L D B^L), \qquad \bar{h}^{\Phi} = z, \tag{25}$$

$$\Lambda^{\Phi,L} = z g_L + D w_L - \tfrac{1}{2}(zB^L D + D B^L z), \tag{26}$$

$$\mathcal{W} = \frac{\partial \Phi}{\partial \gamma} = \sum_L w_L B^L_{aa} - \tfrac{1}{2}\sum_L (B^L z B^L)_{aa}, \tag{27}$$

$$y^c_r = w_r \sum_{tu} (\mathcal{W} + \mathcal{W}^{\mathsf{T}})_{tu}\, E_{tu} c_r, \tag{28}$$

the last one sigma-vector application of a one-body operator, batched over the roots. y^c is raw (not projected), as y is.

6.4 Lagrangian, Z-vector, gate and nuclear term

On a converged single-state CASSCF,

$$\mathcal{L} = E_{\mathrm{CAS}} + E_{\mathrm{corr}} + Z \cdot g_{\mathrm{orb}} + z \cdot g_{\mathrm{ci}} + \langle z^c, f\rangle, \qquad g_{\mathrm{ci}} = 2(H_0 - E)c, \tag{29}$$

g_{orb} `response::fock`'s orbital gradient over the packed non-redundant pairs. Stationarity in the CASSCF parameters $x = (\kappa_{\mathrm{nr}}, c)$ is the Z-vector equation [13]: with $J = \partial(g_{\mathrm{orb}}, g_{\mathrm{ci}})/\partial x$, the CASSCF electronic Hessian, symmetric at a stationary point,

$$J \begin{pmatrix} Z \\ z \end{pmatrix} = -\begin{pmatrix} A^{\mathrm{corr}}_{\mathrm{nr}} + A^{z^c}_{\mathrm{nr}} \\ P_c(y + y^c) \end{pmatrix}, \qquad z \perp c, \tag{30}$$

E_{CAS} contributing no right-hand side because $g = 0$. Its blocks are what the tree has: $J_{\kappa\kappa}$ `response::Hessian`; $J_{c\kappa} v = 2P_c(H[v] - \langle H[v]\rangle)c$, $H[v]$ the active Hamiltonian at the one-index-rotated integrals; $J_{\kappa c} v$ the orbital gradient at the transition RDMs between v and c; and $J_{cc} v = 2P_c(H - E_{\mathrm{CAS}})P_c v$, one sigma vector. It is solved by `response::zvector` (preconditioned CG on

`recipes::pcg`, CI block projected off c before and after the preconditioner, since c is an exact null vector of J_{cc}): orbital right-hand side $g + \mathrm{pack}(A^{\mathrm{corr}} + A^{z^c})$ and CI right-hand side $\frac{1}{2}P_c(y + y^c)$. The $\frac{1}{2}$ is `zvector`'s convention: its CI row is half of Eq. (30)'s, so a caller of Eq. (30)'s form scales its CI right-hand side to match.

The total adjoint. Every term of Eq. (29) but V_{nn} is a function of (h, B) at fixed (c, Z, z, z^c), so the nuclear term reads the sum of their adjoints,

$$\bar{h}^{\mathcal{L}} = \bar{h}^{\mathrm{cas}} + \bar{h}^{\mathrm{corr}} + \bar{h}^{Z} + \bar{h}^{z} + \bar{h}^{z^c}, \qquad \Lambda^{\mathcal{L}} = \Lambda^{\mathrm{cas}} + \Lambda^{\mathrm{corr}} + \Lambda^{Z} + \Lambda^{z} + \Lambda^{z^c} : \tag{31}$$

CAS, $\bar{h}^{\mathrm{cas}} = D$ and $\Lambda^{\mathrm{cas},L}_{pq} = \sum_{rs} \Gamma_{pqrs} B^L_{rs}$ (the core parts J/K shapes, the active part $\Gamma^{(2)}$ against B_{aa}); Z, E_{CAS}'s adjoints at the rotated densities $D^Z = Z^{\mathsf{T}}D + DZ$ and their four-index analogue; z, E_{CAS}'s at the symmetrised transition densities between z and c, times 2; z^c, Φ's above; corr, the reverse pass.

The gate: a free, exact check of the whole solve. $\mathcal{L}$ is stationary in every rotation, so by Eq. (22) its total $X^{\mathcal{L}}$ must be symmetric in every block, gauge blocks included. Its antisymmetric part,

$$A^{\mathcal{L}} = 2(F - F^{\mathsf{T}})(\gamma + \gamma_z, \Gamma + \Gamma_z) + A^{Z} + A^{\mathrm{corr}} + A^{z^c}, \tag{32}$$

with F the generalised Fock at the root's RDMs plus z's transition RDMs, and $A^Z = 2(F' - F'^{\mathsf{T}})$, F' the derivative of F along $C(1 + \epsilon\kappa(Z))$, is evaluated and every block past 10^{-8} is refused, naming it. F is a polynomial of degree 4 in C, so a five-point stencil gives F' exactly to round-off. On the non-redundant pairs $A^{\mathcal{L}}$ is the Z-vector's residual, on the inactive pairs Eq. (24)'s, on the active pairs the gauge identity's: a wrong factor in any term names its block.

The nuclear term. Along any orthonormality-preserving path the symmetric part of θ is $-\frac{1}{2}C^{\mathsf{T}}S^A C$, so with the total adjoints $(\bar{h}^{\mathcal{L}}, \Lambda^{\mathcal{L}}, X^{\mathcal{L}})$,

$$\frac{dE}{dR_A} = \sum_{\mu\nu} \mathcal{P}_{\mu\nu} h^A_{\mu\nu} + \sum_{\mu\nu P} \mathcal{G}^P_{\mu\nu} (\mu\nu|P)^A - \tfrac{1}{2} \sum_{PQ} \mathcal{V}_{PQ} (P|Q)^A - \sum_{\mu\nu} \mathcal{W}_{\mu\nu} S^A_{\mu\nu} + V^A_{nn}, \tag{33}$$

$\mathcal{P} = C\bar{h}^{\mathcal{L}}C^{\mathsf{T}}$, $\mathcal{G}^P = \sum_L (L^{-\mathsf{T}})_{PL} C\Lambda^{\mathcal{L},L}C^{\mathsf{T}}$, $\mathcal{V} = L^{-\mathsf{T}}\mathcal{G}^{\mathrm{aux}}L^{-1}$ with $\mathcal{G}^{\mathrm{aux}}_{LL'} = \langle \Lambda^{\mathcal{L},L}, B^{L'} \rangle$, and $\mathcal{W} = \frac{1}{2}C\,\mathrm{sym}(X^{\mathcal{L}})C^{\mathsf{T}}$. In the code this is one `mcscf::df_gradient` pass: E_{CAS}'s and the multipliers' terms are the ones it builds, and $(\bar{h}, \Lambda, X)$ of $E_{\mathrm{corr}} + \Phi$ are its "extra" MO adjoint. No relaxed density is formed separately.

Algorithm 2 The DF-NEVPT2 nuclear gradient (`sc_gradient`, `pc_gradient`)

Input: converged single-state DF-CASSCF (C, c), core and virtuals canonical; SC or PC
Output: dE/dR

1: $(E_{\mathrm{corr}}, \bar{h}, \Lambda, y, X, A, \varepsilon) \leftarrow$ `sc_adjoint` or `pc_adjoint`
2: $z^c \leftarrow$ Eq. (24); refuse an undetermined degenerate pair
3: $(\Phi, \bar{h}^{\Phi}, \Lambda^{\Phi}, y^c, X^{\Phi}, A^{\Phi}) \leftarrow$ `zc_adjoint`; add field by field
4: $(Z, z) \leftarrow$ `zvector`$\big(g + \mathrm{pack}\,A,\ \frac{1}{2}(y + y^c)\big)$
5: **gate:** $A^{\mathcal{L}}$ below 10^{-8} in all six blocks
6: $dE/dR \leftarrow$ `df_gradient`$\big(\text{multipliers } (Z, z),\ \text{extra } (\bar{h}, \Lambda, X)\big)$

7 The single-state nuclear Hessian

`hessian::df_sc_nevpt2` and `hessian::df_pc_nevpt2` (#816, #817) return $d^2(E_{\mathrm{CAS}} + E_{\mathrm{corr}})/dR_A\,dR_B$ on a converged single-state DF-CASSCF, in k coordinates $q_i = \sum_x Q_{ix}R_x$ (orthonormal rows, the Cartesian ones when none are given). They are all-electron, and a state average or a frozen core is refused. They are assembled in `hessian` from this layer's tangents and Hessian-vector products and from the CASSCF Hessian's parts, with no integrals of their own. There is no QD-NEVPT2 Hessian.

7.1 The $2n+1$ form

Collect the pieces of Eq. (29): the parameters $x = (\kappa_{\mathrm{nr}}, \kappa_{oo}, c)$, the non-redundant CASSCF rotations, the core–core and virtual–virtual rotations the pseudocanonical condition pins, and the CI vector; the multipliers $\lambda = (Z, z, z^c)$; the constraints $g = (g_{\mathrm{orb}}, g_{\mathrm{ci}}, f_{oo})$. At the solution $\mathcal{L}_x = 0$ and $g = 0$, so $dE/dR_A = \mathcal{L}_A$ at fixed (x, λ) along an orthonormal path. Differentiating once more and eliminating λ_B with the two first-order conditions, $g_A + g_x x_A = 0$ and $\mathcal{L}_{xB} + \mathcal{L}_{xx}x_B + g_x^{\mathsf{T}}\lambda_B = 0$,

$$\frac{d^2E}{dR_A\,dR_B} = \mathcal{L}_{AB} + \mathcal{L}_{Ax}\,x_B + x_A^{\mathsf{T}}\mathcal{L}_{xB} + x_A^{\mathsf{T}}\,\mathcal{L}_{xx}\,x_B, \qquad g_x\,x_A = -g_A. \tag{34}$$

The parameters are needed to first order, one linear solve per coordinate. The multipliers are the gradient's own, at zeroth order, so their $3N$ responses never appear, and no second-order response x_{AB} does either. SC's amplitudes are closed form and PC's come from a solve whose factorisation the tangent reuses, so neither needs an amplitude response of its own.

7.2 The parameter response and the tangent

g_x is block triangular: the CASSCF constraints transform covariantly under κ_{oo} and do not respond to it at a stationary point.

1. **CP-CASSCF**, $J(\kappa_{\mathrm{nr}}, c)_A = -\partial g_{\mathrm{CAS}}/\partial R_A$, J the gradient's CI-relaxed Hessian: `hessian::df_casscf`'s step (or `df_casscf_cp` alone), `response::zvectors` on all $3N$ columns at once, CI right-hand sides halved to the Z-vector's convention. A state-specific excited root is a saddle in (κ, c) and is refused (Divergence); so is a singular $\tilde{H}$, detected by a probe column (#978).

2. **The pseudocanonical block**, closed form once 1. is done:

$$\kappa^A_{pq} = -\frac{f^{(A)}_{pq}}{\varepsilon_p - \varepsilon_q}, \qquad (p,q) \text{ core–core or virtual–virtual}, \tag{35}$$

   $f^{(A)}$ the total first-order change of f at fixed κ_{oo}. Below $|\varepsilon_p - \varepsilon_q| = 10^{-6}$, $\kappa^A_{pq} = 0$ while $|f^{(A)}_{pq}| \le 10^{-8}$ (a split the displacement does not lift); past that SC refuses, naming the pair and the coordinate, and PC records a degenerate pair, which its Hessian refuses (§7.5).

Eq. (35) is z^c's division again, now in a response that enters Eq. (34) quadratically. PC does not escape it: this tree's PC puts each tuple at its own Δ_t from f's diagonal and is not invariant to inactive rotations, so κ_{oo} stays in PC's Hessian too.

`hessian::df_nevpt2_tangents` then forms the total first-order change of the NEVPT2 functional's inputs along each coordinate, in the MO basis:

$$\dot{h}^x = h^x + \kappa^{x\mathsf{T}}h + h\kappa^x, \qquad \dot{B}^x_Q = W^x_Q + \kappa^{x\mathsf{T}}B_Q + B_Q\kappa^x, \qquad \dot{c}^x = c^x, \qquad \kappa^x = \kappa^x_{\mathrm{nr}} + \kappa^x_{oo}, \tag{36}$$

(h^x, W^x) the derivative integrals along the Löwdin connection $\tau^x = -\frac{1}{2}S^x_{\mathrm{MO}}$ (`mcscf::df_integral_derivatives`), W^x's fitted index carried in the gauge $\mu^x = -\frac{1}{2}m^x$, $m^x = L^{-1}M^xL^{-\mathsf{T}}$. $f^{(x)}$ comes from `nevpt2::builders_tangent`, the forward tangent of every builder the energy reads, each term by its product rule over the forward's own builds. The RDMs' tangent is exact in three `rdms` calls: the bra/ket builds at $(\dot{c}, c)$ and $(c, \dot{c})$, and the one-vector build over $(tu|vw)$'s tangent, of which only f3 is kept.

7.3 The Hessian-vector product

`sc_hessian_product` and `pc_hessian_product` (#814, #815) are the tangent of `*_adjoint` + `zc_adjoint` at fixed z^c along directions $(\dot{h}, \dot{B}, \dot{c})$, so $\langle t_A, \mathrm{HVP}\, t_B\rangle$ is the $E_{\mathrm{corr}} + \Phi$ share of Eq. (34) along Eq. (36)'s t. The adjoint is $J^{\mathsf{T}}\bar{E}$, $\bar{E}$ the class walk's bars at the builders' outputs and J the builders' Jacobian, so its tangent is forward over reverse, two kinds of term, one sweep (`bars_tangent`):

- $J^{\mathsf{T}}\dot{\bar{E}}$: the class walk's bars along the builders' tangent (`class_energies_hvp`, `pc_class_energies_hvp`), run back through the gradient's own reverse pass;
- $\dot{J}^{\mathsf{T}}\bar{E}$: every builder term's adjoint at the fixed bars with one other operand at its tangent, the same einsum with the operand swapped (`adjoint_slot`), so no derivative is written by hand.

SC's class tangent. The multipliers' tangent is the finish's 3×3 Hessian. In the (N, d) form, $d = N\Delta + H$ and $E = -N^2/d$, which carries no removable $1/N$:

$$\begin{aligned} E_{NN} &= -\frac{2}{d} + \frac{4N\Delta}{d^2} - \frac{2N^2\Delta^2}{d^3}, & E_{NH} &= \frac{2N}{d^2} - \frac{2N^2\Delta}{d^3}, & E_{N\Delta} &= \frac{3N^2}{d^2} - \frac{2N^3\Delta}{d^3}, \\ E_{HH} &= -\frac{2N^2}{d^3}, & E_{H\Delta} &= -\frac{2N^3}{d^3}, & E_{\Delta\Delta} &= -\frac{2N^4}{d^3}, \end{aligned} \tag{37}$$

whose (H, Δ) block has rank one, and whose degree-one homogeneity in (N, H) gives $NE_{NN} + HE_{NH} = 0 = NE_{NH} + HE_{HH}$. Times the forward tangent's $(\dot{N}, \dot{H}, \dot{\Delta})$ it gives $(\dot{P}, \dot{Q}, \dot{R})$, taken onto every bar by one reverse pass at fixed multipliers. The bars at fixed multipliers are themselves polynomials of degree at most five along a line (two slabs, each quadratic in B, around a block of degree at most two), so

$$\sum_{k=1}^{3} w_k \, \frac{A(x + ks\dot{x}) - A(x - ks\dot{x})}{2ks}, \qquad w = (\tfrac{3}{2}, -\tfrac{3}{5}, \tfrac{1}{10}), \tag{38}$$

is their derivative exactly, the weights cancelling the s^2 and s^4 terms. A one-function class reads $\dot{\eta}$ from the blocks' polarisation.

PC's class tangent. The same split over the solve. With $g = y/(\lambda + \Delta)$ the multipliers move as $\dot{g} = (\dot{y} - \dot{D}g)/(\lambda + \Delta)$, $\dot{D} = \dot{K}' + \Delta\dot{S}' + \dot{\Delta}$, the solve differentiated through its own factorisation. U's tangent $V_n\Theta'$ enters through a symmetric difference that is exact (the bars are quadratic in U), the inputs' tangent at fixed multipliers through the same three-point formula, and the kept span's rotation through the Daleckii–Krein derivative of the span bar. Every divided difference there is a product of two first-order ones, so no pair on one side of the cut is ever divided by. These are the null-space rotation terms of $S^+ = UU^{\mathsf{T}}$: zero for exact null modes at an eigenstate, and of the first-order term's size for real modes near the cut.

Φ, and the functional of c. $\Phi = \langle z^c, f\rangle$ is linear in h, bilinear in B and quadratic in c, so its share at fixed z^c is closed form. Two conventions fix the CI–CI block, and the gradient shares both. The product is the Hessian of the *Rayleigh-normalised* functional $F(h, B, c) = (E_{\text{corr}} + \Phi)(h, B, c/|c|)$, so the sphere's $-(u \cdot v)\,(\partial\mathcal{L}/\partial c) \cdot c$ is included: with $P = 1 - cc^{\mathsf{T}}$ and $u = \dot{c} - (c \cdot \dot{c})c$, the c-gradient's tangent is $P\dot{y}(\dot{h}, \dot{B}, u) - u\,(c \cdot y) - c\,(u \cdot y) - (c \cdot \dot{c})Py$. The off-shell extension is the rank-reduced one, $\hat{H}_D|\Psi_0\rangle = E_0|\Psi_0\rangle$ off the root too, with no Kollmar residual [12]. A residual-including CI–CI block against this gradient's z, or the reverse, would be wrong with no symptom.

7.4 Assembly

Every part is the second derivative of its term of $\mathcal{L}$ along one path: the orbitals $C(R)\exp K$ along the Löwdin connection $C(R) = C(C^{\mathsf{T}}S(R)C)^{-1/2}$, $K = \kappa_{\text{nr}} + \kappa_{oo}$, the CI vector $\exp(A)c$ on the unit sphere, and B's fitted index in one gauge. A part in another parametrization would leave a term of second order in κ unaccounted. With $N = E_{\text{corr}} + \Phi$ a function of $I = (h, B)$ and c, $O(M, \kappa) = \kappa^{\mathsf{T}}M + M\kappa$ and $t_A = I^A + O(I, \kappa^A) + c^A$:

$$\begin{aligned} N^{\text{total}}_{AB} = {} & \langle N_I, I^{AB}\rangle + \langle t_A, N''\, t_B\rangle \\ & + \langle N_I, O(I^A, \kappa^B) + O(I^B, \kappa^A)\rangle + \langle N_I, \tfrac{1}{2}O(O(I, \kappa^A), \kappa^B) + (A \leftrightarrow B)\rangle, \end{aligned} \tag{39}$$

the skeleton, the product, and the rotation cross terms $\tau\kappa$ and $\kappa\kappa$. For symmetric M, $\langle N_I, O(M, \kappa)\rangle = \langle \kappa, X(M)\rangle$, $X(M) = 2(M_h\bar{h} + \sum_Q M_Q\Lambda_Q)$, so the cross terms are n^2_{mo} Gram matrices. The parts and what computes each:

part	computed by
E_{CAS}	`hessian::df_casscf` as is; its κ_{oo} rows vanish by $E_{\text{CAS}}(CU, c) = E_{\text{CAS}}(C, c)$
t_x	`df_nevpt2_tangents` over its λ^x, with κ^x_{oo}
$(\bar{h}, \Lambda)$, multipliers	`*_adjoint`, `zc_multipliers`, `zc_adjoint`, `lagrangian_gradient`: the gradient's chain, adjoint kept
$\langle t_A, N''t_B\rangle$	`*_hessian_product` along each coordinate's t, paired with every t_A by three GEMVs
Z terms	`df_casscf_z`: $Z \cdot \partial^2 g/\partial R\,\partial R$, $I^x Z$ against λ^y both ways, and $\lambda^{\mathsf{T}}(Z \cdot \partial^2 g)\lambda$, the CASSCF third derivative
skeleton	`df_adjoint_xy`: $\sum \bar{h}\, h^{xy} + \sum \Lambda\, B^{xy}$
$\tau\kappa$, $\kappa\kappa$	X Gram matrices from n^2_{mo} stacks
κ_{oo} rows	$\frac{1}{2}\big(\langle \kappa^B_{oo}, [X_N, \kappa^A_{\text{nr}}]\rangle + (A \leftrightarrow B)\big)$

The κ_{oo} rows follow from BCH to second order, $\exp(sP{+}tQ) = \exp(sP)\exp(tQ)\exp(-\frac{1}{2}st[P, Q]){+}O(3)$, and the covariance of the orbital gradient: of $Z \cdot g$'s second derivative only $-\frac{1}{2}\nabla(Z \cdot g) \cdot ([\kappa^A_{\text{nr}}, \kappa^B_{oo}] + [\kappa^B_{\text{nr}}, \kappa^A_{oo}])$ survives, and the Z-vector equation turns $\nabla(Z \cdot g)$ into $-X_N$. N's own κ_{oo} terms are in t's κ, so the product and the cross terms carry them.

The skeleton's gauge. h_{MO} and B_{MO} are differentiated along $\tau^{xy} = -\frac{1}{2}S^{xy} + \frac{3}{8}\{S^x, S^y\}$ and, on the fitted index, $\mu^{xy} = -\frac{1}{2}m^{xy} + \frac{3}{8}\{m^x, m^y\}$: the symmetric square root's second order, whose first order is exactly W^x's gauge. The fitted index is defined only up to an orthogonal turn, which a general Λ sees; the product's term pairs W^x and this one B^{xy}, and their sum is the derivative of one function only because both are the same curve.

The result is exactly symmetric; the antisymmetric part discarded and the translational sum are returned.

7.5 Where there is no Hessian: the refusals

Each refusal below is Numerical, names what failed, and returns no Hessian; the CP solve's own (a saddle, a singular $\tilde{H}$) are passed on as they come.

SC: empty perturbers. At a symmetric geometry a perturber can be empty by symmetry ($|N| \le 10^{-14}$); a direction that breaks the symmetry populates it at second order, where its energy is homogeneous of degree two in the direction. Along one direction u it is a quadratic form, and the perturber takes the demo's K6, the finish's multipliers at the ray's limit $\eta = H^{uu}/N^{uu}$. Along a pair (u, v) that both populate it, it takes K6 at the mean η and is scored by $\rho_\tau = \max(|\eta_u - \eta_v|, 2|\kappa - c\eta|)/|\Delta + \eta|$ at unit populations, zero exactly when $H \propto N$ on the pair, weighted by the perturber's share of the pair's Hessian (#914). Past 10^{-6} the pair is refused: the energy is a second derivative along every line but no bilinear form, #894's SC kink. Because the product is not linear in the direction there, the Hessian is taken along each coordinate's own t and pairs of symmetry-breaking coordinates are taken again as each other's partners. A coordinate breaks the symmetry when its population exceeds 10^{-10} (`kEmptyPopulationFloor`) of the *largest* scale over the coordinates (#952: a rigid translation's tangent is round-off and populates everything against its own scale). At `nh3_cs_sto3g` in symmetry-adapted coordinates the whole A″×A″ off-diagonal block is refused, which is the right output.

SC: mixing inactive pairs (#952). SC is not invariant to inactive rotations. Where a coordinate couples two near-degenerate inactive orbitals, the pair's 2×2 block $[[\varepsilon_p, Ft], [Ft, \varepsilon_q]]$ has gap $\sqrt{g^2 + 4F^2t^2}$, and its avoided crossing has half-width $t_{\mathrm{mix}} = g/2|F| = 1/(2|\kappa^i_{oo,pq}|)$. Past it the pair has rotated and the curvature is a different branch's. At `examples/nevpt2.toml`'s C_{2v} water, virtuals $1.4\,\mathrm{m}E_h$ apart give $t_{\mathrm{mix}} = 5.6 \times 10^{-4}$ bohr along B_2, and SC's B_2 frequency is 14063 against PC's $4028\,\mathrm{cm}^{-1}$. The analytic element is the correct $t \to 0$ limit, which a ladder of the gradient reaches only below 10^{-4} bohr. Any coordinate with $t_{\mathrm{mix}} < 10^{-3}$ bohr is refused, naming the pair and recommending PC.

PC: no partners, but the Schur lift. PC has no empty perturbers: a tuple that vanishes by symmetry has $b = Sx = 0$ and moves as a smooth quadratic. Instead a null mode v of S can rise at second order along the CI vector's path. With $S(t) = S + tS' + \frac{1}{2}t^2S''$ the null block's eigenvalues rise as $\frac{1}{2}t^2\,\mathrm{eig}(L)$,

$$L = \mathcal{P}\,(S'' - 2S'S^{+}S')\,\mathcal{P}, \tag{40}$$

$\mathcal{P}$ the null projector, $L \succeq 0$. Structural null modes never lift ($L = 0$), spin ones not along singlet directions, and symmetry ones do along a singlet that breaks the symmetry. A lifted mode crosses the cut at $t^* = \sqrt{2\,\mathrm{cut}\,\sigma_{\max}/\ell}$, $\ell = \mathrm{eig}(L)$, beyond which the energy carries its share. The product stays the exact derivative of the dropped branch and reports t^* per direction and class. Along the coordinate's own t (its CI part c^B) t^* is in bohr, and any coordinate and class with $t^* < 10^{-3}$ bohr is refused. HCN's CAS(6,6) with its σ pair active lifts Sir's modes at $2\text{–}5 \times 10^{-4}$ bohr on every bend; at $f = 10^{-10}$, t^* grows to 5×10^{-3}–10^{-2} and nothing is refused.

PC: thin kept spectra and degenerate pairs. A class whose `kept` is below 10^{-10} of its $\sigma_{\max}$ is refused, naming the floor that would cut it (#917; #943 reaches it at a C_{2v} formaldehyde CAS(6,6)). A degenerate inactive pair that a coordinate splits (gap below $10^{-6}\,E_h$, $|f^{(x)}_{pq}| > 10^{-8}$) is refused, since κ^x_{oo} is not a division the state determines there. The check is conservative: any Cartesian record refuses, even if the caller's coordinates combine them so that f vanishes.

PC near a degenerate pair, measured. `h2co_neardeg_sto3g`'s virtual pair is $6 \times 10^{-3}\,E_h$ apart, $\max|\kappa^x_{oo}| = 21.8$, and its reference ladder runs through gaps of 5.6×10^{-5}. The terms that grow as $1/\mathrm{gap}^2$, $\kappa\kappa$ and the product's κ_{oo}–κ_{oo} block, are each about 10^9 bars and cancel to the fixture's bar: about nine of the parts' sixteen digits go into it. So PC's second-order κ_{oo} terms do cancel the $1/(\varepsilon_p - \varepsilon_q)$ of Eq. (35), at the cost of digits, and that pair is not refused.

7.6 The semi-numerical Hessian and the bilinearity check

The numerical alternative differences the analytic gradient,

$$H_{AB} \approx \frac{g_A(R + s\,q_B) - g_A(R - s\,q_B)}{2s}, \qquad H \leftarrow \tfrac{1}{2}(H + H^{\mathsf{T}}), \tag{41}$$

along each coordinate q_B, in the pipeline by a fourth-order stencil. Each displaced CASSCF starts from the reference's orbitals orthonormalised in the displaced metric, $C(S_{\mathrm{MO}})^{-1/2}$, and its CI, and passes the CASSCF chain's continuity guards. It is also the oracle for the analytic Hessian. It does not escape the differentiability problem of §7.5: along a pair of directions that leave the symmetry, a difference converges, if at all, to a polarisation of directional second derivatives that depends on the step and on the combination of coordinates.

The diagnostic. For coordinates q_1, q_2, check

$$H(q_1 + q_2,\ q_1 + q_2) \stackrel{?}{=} H(q_1, q_1) + H(q_2, q_2) + 2H(q_1, q_2), \tag{42}$$

with the step dependence of each element. A violation above the difference noise means that block has no Hessian, and it is reported rather than published as a frequency. With one coordinate in an irrep Eq. (42) is empty and the ladder is the only evidence.

Which coordinates. Across two irreps the cross block vanishes by symmetry in symmetry-adapted coordinates, but the function need not be a quadratic form there either. A Cartesian step mixes irreps, so a Cartesian difference Hessian at a symmetric point depends on the frame, silently. Symmetry-adapted coordinates avoid that and name the failing block; on Cartesian pairs the check still detects a failure, at two more gradients per pair, without naming the irrep. The tree detects no point group, so the coordinates are the caller's (`[hessian] coordinates`, orthonormal rows). Element (i, j) is $q_i^{\mathsf{T}}\,\partial g/\partial q_j$, and a Cartesian Hessian is $Q^{\mathsf{T}}HQ$.

In the pipeline (#812). Under `[nevpt2_gradient]` the numerical `[hessian]` runs a ladder by default: each column at $s = h$, $h/2$, $h/4$, the ratio of successive changes judged in $(12.8, 19.2)$ where the finer change is past ten noise floors, the floor measured from restarts and from resolves at tolerances a hundred times tighter. It checks Eq. (42) on every coordinate pair up to twelve coordinates. A failing pair withholds the frequencies: the run fails, naming the pairs. At `nh3_cs_sto3g` under SC in symmetry-adapted coordinates all ten $\mathrm{A''}\times\mathrm{A''}$ pairs fail and the cross-irrep pairs pass, which is the block the analytic route refuses.

7.7 In the pipeline

`[hessian]` beside a single-state `[nevpt2_gradient]` is that variant's $E_{\mathrm{CAS}} + E_{\mathrm{corr}}$ Hessian (#818): analytic by default, or central differences of `sc_gradient`/`pc_gradient` over $6N$ displaced geometries under a Richardson ladder with a bilinearity check (#812). `coordinates`, when given,

are passed through and the result reported in Cartesians, $Q^{\mathsf{T}}HQ$: at a symmetric geometry give the symmetry-adapted ones. Under `variant = "pc"` the whole chain runs at `pc_floor` $= 10^{-10}$ (`HessianStep::kPcFloor`), the outer `[nevpt2_gradient]` included, so the gradient written beside the Hessian is the same energy's; `pc_floor = 0` restores the round-off cut bitwise. Refused before anything runs: integral-direct ERIs, a state average, `quasi_degenerate`, a frozen core and a `[casscf_gradient]` beside it. Stationarity is assumed, not checked: converge `[casscf]` to `conv_tol_grad` $= 10^{-10}$. Harmonic analysis follows; there are no IR intensities.

8 The QD-NEVPT2 gradient

8.1 The functional $\mathcal{E}[W]$

For any symmetric $n \times n$ W,

$$\mathcal{E}[W] = \sum_{MN} W_{MN} H^{\text{eff}}_{MN} = \sum_{M} \rho_M E_M + \sum_{MN} W_{MN} C_{MN}, \qquad \rho_M = W_{MM}, \tag{43}$$

and state K's energy is $E_K = \mathcal{E}[u_K u_K^{\mathsf{T}}]$ ($u_K = R_K$; Park's R_{PPMN} [10]). R_K is an eigenvector of a symmetric matrix, so $\delta E_K = u_K^{\mathsf{T}} \delta H^{\text{eff}} u_K$: R needs no multiplier, and the gradient of E_K is that of $\mathcal{E}[W]$ at fixed W. Against a symmetric W the symmetrisation drops out, so the unsymmetrised couplings are what is differentiated. `qd_sc_energy` and `qd_pc_energy` build every input from (h, B, C) and the roots at their SA weights: one set of shared arrays, ε from f^{SA} with γ^{SA} rebuilt from the roots at every point, and each ordered pair's dm1–dm3.

The reverse pass. `qd_*_adjoint` runs the QD reverse pass at $\bar{H} = W$, hands each transition's dm1–dm3 bars (folded into dm3 along Eq. (13)) to the bra/ket `rdms_adjoint` with null f3 bars, then the shared builders backwards, so that $\bar{\gamma}^{\text{SA}}$ collects in one bar. Each root then takes $w_r \bar{\gamma}^{\text{SA}}$, whatever its W_{rr}: every root carries ε's share through γ^{SA}.

8.2 The Lagrangian on a state average

The SA-CASSCF parameters are the non-redundant rotations, the roots c_r and the rotations Θ among them. With (Z, z, Θ) in `response::zvector`'s coordinates,

$$\mathcal{L} = \mathcal{E}[W] + Z \cdot g_{\text{SA}} + \sum_r 2 w_r \, z_r \cdot (H - E_r) c_r + \sum_{r<s} \big(2(w_r - w_s)\Theta_{sr} + \zeta_{sr}\big) \langle c_s | H | c_r \rangle + \langle z^c, f^{\text{SA}} \rangle, \tag{44}$$

Park's Eq. 24 [10] made explicit about the rotations among the roots: a pair of unequal weights is a row of the SA Hessian (Θ), and a pair of equal weights, to which E_{SA} is blind, needs a multiplier ζ of its own. Unlike a CASSCF root energy, C_{MN} is not stationary under mixing the references, which is QD-NEVPT2's reference-basis dependence [7]. Along $c_r \to c_r + \theta c_s$, $c_s \to c_s - \theta c_r$,

$$\tau_{sr} = c_s^{\mathsf{T}} \bar{y}_r - c_r^{\mathsf{T}} \bar{y}_s, \qquad \zeta_{sr} = -\frac{\tau_{sr}}{E_s - E_r} \ (w_r = w_s), \qquad t_{sr} = \frac{\tau_{sr}}{2(w_r - w_s)} \ (w_r \neq w_s), \tag{45}$$

$\bar{y}_r$ the raw CI bar of $\mathcal{E}^{\text{corr}}[W] + \Phi^{\text{SA}}$. ζ's orbital term is $\zeta_{sr} g^{(sr)}$, the orbital gradient of $\langle c_s | H | c_r \rangle$ (`response::fock` at the transition RDMs less at zero RDMs), and its nuclear term is the Hellmann–Feynman gradient of $\langle c_s | H | c_r \rangle$, folded over all pairs into one set $(\gamma^{\zeta}, \Gamma^{\zeta})$.

The Z-vector. Right-hand sides: orbital $g^\rho + \sum \zeta_{sr} g^{(sr)} + \text{pack}\, A$, g^ρ the Fock's orbital gradient at the ρ-weighted RDMs; CI $\bar{y}_r/(2w_r)$ for every root (`zvector`'s CI rows are the SA Hessian's divided by $2w_r$); rotations t. At $n = 1$ this is §6's $\frac{1}{2}(y + y^c)$. A root of weight below the rotation tolerance is refused, since it has no SA CI row. The gate is the single-state one at the ρ-weighted RDMs plus ζ's and z's.

Two derivative passes.

$$\frac{dE_K}{dR} = \underbrace{\texttt{df_gradient}(\rho;\ (Z, z, \Theta);\ \text{extra})}_{\text{reference at } \rho,\ \text{multipliers},\ \mathcal{E}^{\text{corr}}+\Phi^{\text{SA}}} + \underbrace{\texttt{df_transition_gradient}(\gamma^\zeta, \Gamma^\zeta)}_{\zeta,\ \text{skipped when } \zeta=0}. \tag{46}$$

8.3 The W form, and several W at once

Everything in $\mathcal{L}$ is linear in W except the core energy, which $\sum_M \rho_M E_M$ carries at $\operatorname{tr} W$. The W form (`qd_*_gradient(..., w)`) takes any symmetric W: $\rho = \operatorname{diag} W$; g^ρ and the gate carry the core part $\operatorname{tr} W$ times; and the first pass is one `df_gradient` at the mix ρ when $\sum \rho = 0$ (its core-off path) or $\sum \rho = 1$ with $\rho \geq 0$. Any other $\rho = (\operatorname{tr} W)p + d$, $p \geq 0$ normalised, takes one unrelaxed pass at p scaled by $\operatorname{tr} W$ and one relaxed pass at d. So $G(W_1 + W_2) = G(W_1) + G(W_2)$ and $\sum_K dE_K/dR = G(1)$. The state form is a wrapper that builds $W = u_K u_K^\mathsf{T}$, refusing a QD state within $10^{-8}\, E_h$ of another.

A list of W's shares the forward arrays, the Fock, the CI-relaxed Hessian and one block Z-vector solve (`response::zvectors`) over all right-hand sides, each column's true residual checked against 10^{-10}. The derivative passes stay one per W: one three-centre derivative kernel is most of each pass, so stacking them would need a multi-density kernel.

Algorithm 3 The DF-QD-NEVPT2 gradient $G(W)$ (`qd_lagrangians` and `qd_passes`)

Input: DF-SA-CASSCF $(C, \{c_r\}, w, \{E_r\})$, core and virtuals canonical over f^{SA}; SC or PC; symmetric W (state form: $W = u_K u_K^\mathsf{T}$ from H^{eff})

Output: $G(W)$; for the state form dE_K/dR

1: $(\bar{h}, \Lambda, \bar{y}_r, X, A, \varepsilon) \leftarrow$ `qd_sc_adjoint` or `qd_pc_adjoint` at W
2: $z^c \leftarrow$ Eq. (24) over A and ε^{SA}; Φ^{SA}'s adjoints at the weights; add
3: τ; ζ on equal-weight pairs, t on unequal ones; ζ's orbital gradient and $(\gamma^\zeta, \Gamma^\zeta)$
4: $\rho \leftarrow \operatorname{diag} W$; g^ρ at the ρ-weighted RDMs, core part at $\operatorname{tr} W$
5: $(Z, z, \Theta) \leftarrow$ `zvector`$(g^\rho + \zeta \cdot g + \text{pack}\, A,\ \bar{y}_r/(2w_r),\ t)$
6: **gate** per block
7: $G \leftarrow$ `df_gradient`$(\rho, (Z, z, \Theta), \text{extra})$ + `df_transition_gradient`$(\gamma^\zeta, \Gamma^\zeta)$

The dropped rank-projector terms (PC). Eq. (17) and its QD form differentiate the restricted inverse as if S_N's kept span were fixed, exact when S_N's dropped modes annihilate $|N\rangle$, at eigenstates. Ladders along each real root c_r are what check it, since random RDM directions carry no eigenstate structure. The residual terms of an unconverged CASCI are the ones Kollmar et al. [12] show reach the 5-RDM in the Dyall matrix elements.

9 Interstate and derivative couplings

9.1 The interstate coupling

With $W^{PQ} = \frac{1}{2}(u_P u_Q^\mathsf{T} + u_Q u_P^\mathsf{T})$ (Park's R_{PQMN}),

$$\mathbf{h}_{PQ} = u_P^\mathsf{T}(\nabla H^{\text{eff}})u_Q = G(W^{PQ}), \qquad \operatorname{tr} W^{PQ} = 0, \tag{47}$$

so the first pass takes `df_gradient`'s core-off path: the reference part $\sum_r R_{rP}R_{rQ}\nabla E_r$ has no core energy and no nuclear repulsion. Since $u_P^\mathsf{T}\delta H^{\text{eff}} u_Q = (E_Q - E_P)u_P^\mathsf{T}\delta u_Q$,

$$\mathbf{d}_{PQ}^{\text{NEV}} = \sum_M R_{MP}\nabla R_{MQ} = \frac{\mathbf{h}_{PQ}}{E_Q - E_P}. \tag{48}$$

A pair within $10^{-8}\,E_h$ is refused. $\mathbf{h}_{PQ}$ carries R_P's sign times R_Q's, and so the roots' phases only through them.

9.2 The derivative coupling

With $\tilde{\Psi}_P = \sum_M(|M\rangle + |\Psi_M^{(1)}\rangle)R_{MP}$, Park's Eqs. 45–50 [10] (after the MS-CASPT2 derivation of Park and Shiozaki [11]) give

$$\mathbf{d}_{PQ} = \mathbf{d}_{PQ}^{\text{NEV}} + \mathbf{d}_{PQ}^{\text{CAS}} + \mathbf{d}_{PQ}^{\text{PT2}}, \qquad T_{MN}^{PQ} = \tfrac{1}{2}(R_{MP}R_{NQ} - R_{MQ}R_{NP}), \tag{49}$$

$$\mathbf{d}_{PQ}^{\text{CAS}} = \sum_{MN} T_{MN}^{PQ}\langle M|\nabla N\rangle, \qquad \mathbf{d}_{PQ}^{\text{PT2}} = \sum_{MN} T_{MN}^{PQ}\langle \Psi_M^{(1)}|\nabla N\rangle = \sum_{xy} D_{xy}^{\text{PT2}}\langle\phi_x|\nabla\phi_y\rangle, \tag{50}$$

the last because $\Psi_M^{(1)}$ lies outside the active space, so ∇ reaches it only through $|N\rangle$'s orbitals, with

$$D_{xy}^{\text{PT2}} = \sum_{MN} T_{MN}\langle\Psi_M^{(1)}|E_{xy}|N\rangle. \tag{51}$$

Both $\mathbf{d}^{\text{CAS}}$ and $\mathbf{d}^{\text{PT2}}$ are one `mcscf::df_nac` combination

$$\mathbf{d}^T = \sum_{MN} T_{MN}\mathbf{d}_{MN} + \mathbf{d}[D^X], \qquad \mathbf{d}[D^X] = \sum_{xy} D_{xy}^X\langle\phi_x|\nabla\phi_y\rangle, \tag{52}$$

with the SA orbitals and CI relaxed: $\mathbf{d}^{\text{CAS}}$ at $(T^{PQ}, D^X = 0)$ and $\mathbf{d}^{\text{PT2}}$ at $(T = 0, D^X = D^{\text{PT2}})$. Its pieces, by linearity in the pair coupling:

- **CI part:** the transition Lagrangian at the combined RDMs $\sum_{M<N} 2T_{MN}/(E_N - E_M)\,(\gamma^{(MN)}, \Gamma^{(MN)})$ (symmetrised), one Z-vector and one `df_transition_gradient`.
- **CSF part:** $\frac{1}{2}G[D^\mathsf{T} - D]$ at $D = C_a\gamma^T C_a^\mathsf{T}$, $\gamma^T = \sum_{MN} T_{MN}\gamma^{(MN)}$, with $G[a]_{A,k} = \sum_{m\in A}\sum_n a_{mn}\,\partial S_{mn}/\partial A_k$ the half-derivative overlap (PySCF's `nac_csf`).
- D^X**:** with $C(R) = C_0(C_0^\mathsf{T}S(R)C_0)^{-1/2}e^\kappa$, the *basis motion* $\frac{1}{2}G[C(D^{X\mathsf{T}} - D^X)C^\mathsf{T}]$, and the *orbital response*, $D_{pq}^X - D_{qp}^X$ added to the Z-vector's orbital right-hand side. D^X has no CI part.

`mult_ediff`	`use_etfs`	`total`
false	false	$\mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\text{CAS}} + \mathbf{d}^{\text{PT2}}$ (default)
false	true	$\mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\text{CAS}} + \mathbf{d}^{\text{PT2}} - \text{csf}$
true	false	$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\text{CAS}} + \mathbf{d}^{\text{PT2}})$
true	true	$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\text{CAS}} + \mathbf{d}^{\text{PT2}} - \text{csf})$

$\Delta E = E_Q - E_P$, and csf is $\mathbf{d}^{\text{CAS}}$'s CSF part plus $\mathbf{d}^{\text{PT2}}$'s basis motion (the terms electron-translation factors drop).

9.3 The PT2 transition density

T is antisymmetric, so only $M \neq N$ enter and every term that reads no RDM carries $\langle M|N\rangle = 0$. $E_{xy}|N\rangle$ lies in the first-order interacting space only for $(x, y) = (t, i)$, (a, t) and (a, i), one class each (Si, Sr, Sir); every other element of D^{PT2} is exactly zero. In that class $E_{xy}|N\rangle$ is the one-body function χ_y^N of V's h_v term, and $\Psi_M^{(1)}$ is a combination of the class's internal functions ϕ_μ^M (and, for SC, of χ^M). So with the T-weighted transition RDMs $\rho^M = \sum_N T_{MN}\gamma^{(MN)}$ (dm1 and dm2, bra M), each block needs only

$$\langle \phi_\mu^M | E_{xy} | \rho^M \rangle = S_1(\rho^M)[\mu, y], \qquad \langle \chi_a^M | E_{xy} | \rho^M \rangle = S_{11}(\rho^M)[a, y], \tag{53}$$

S_1 the class's PC one-body metric and S_{11} SC's $h_v \cdot h_v$ metric, both over ρ^M. No dm3, no f3, no 4-RDM.

class	block	$S_1(\rho)$	$S_{11}(\rho)$
Sr, over r	$D[r, u]$	$\rho_2[r', p, q, u]$, $\mu = (p, q, r')$	$\rho_1[a, u]$
Si, over i	$D[u, i]$	dm2_h$(\rho)[r, p, q, u]$	$-\rho_1[u, a]$
Sir, over (r, i)	$D[r, i]$	$[2\rho_1[q, p]; -\rho_1[q, p]]$, $\mu = (x, p, q)$	0

with dm2_h$(\rho)[a, b, c, d] = 2\rho_1[a, b]\delta_{cd} - \rho_2[a, b, d, c]$.

SC. With the handed-over c_t^M and slab rows x,

$$D[r, u] = \sum_M c_r^M \Big(\sum_{pqr'} x[r, p, q, r']\, \rho_2^M[r', p, q, u] + \sum_a (h_v^{\mathrm{Sr}})_{ra}\, \rho_1^M[a, u] \Big), \tag{54}$$

$$D[u, i] = \sum_M c_i^M \Big(\sum_{pqr} x[q, p, i, r]\, \mathrm{dm2_h}(\rho^M)[r, p, q, u] - \sum_a (h_v^{\mathrm{Si}})_{ai}\, \rho_1^M[u, a] \Big), \tag{55}$$

$$D[r, i] = \sum_M c_{ri}^M \sum_{pq} \big(2x_1[r, p, i, q] - x_2[r, p, q, i]\big) \rho_1^M[q, p], \tag{56}$$

one more walk of the Sr, Si, Sir1 and Sir2 slabs.

PC. $\Psi_M^{(1)} \ni -\sum_k g_{tk}^M \sum_\mu U_M[\mu, k]\phi_\mu$ has no χ part, so with $W^M = S_1(\rho^M)^{\mathsf{T}} U_M$, $D[\cdot, t] = -\sum_M \sum_k W^M[\cdot, k]\, g_{tk}^M$: two einsums per state and class, no slab.

9.4 Three solves or one: the fold

The default path solves three Z-vectors per pair ($\mathbf{h}_{PQ}$'s, and the combination twice) so that $\mathbf{h}_{PQ}$, $\mathbf{d}^{\mathrm{CAS}}$ and $\mathbf{d}^{\mathrm{PT2}}$ can be reported apart. Every right-hand side is linear in its source and all three share the CI-relaxed SA Hessian, so Park's Eqs. 51–52 [10] fold them:

$$\mathbf{h}_{PQ} + \Delta E\big(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}} - \mathrm{csf}\big) = G(W^{PQ})\Big|_{\substack{\mathrm{RDMs}+\Delta E(\gamma,\Gamma)_{T^{PQ}} \\ A+\Delta E(D^{\mathrm{PT2}} - D^{\mathrm{PT2T}})}}, \tag{57}$$

one Z-vector whose orbital right-hand side and Hellmann–Feynman pass take ΔE times the combination's RDMs, and whose A takes ΔE times D^{PT2}'s antisymmetric part; csf needs no Z-vector. The fold's RDMs are no state's, so their own active–active asymmetry (the CSF term's, which no multiplier removes) is taken out of the gate in that block. It costs about 0.55–0.68 of the three-solve path, and is off by default because it cannot report the three parts separately.

Algorithm 4 The QD-NEVPT2 derivative coupling (`qd_sc_nac`, `qd_pc_nac`)

Input: DF-SA-CASSCF; SC or PC; QD states $P \neq Q$; `NacOptions`; fold
Output: $\mathbf{d}_{PQ}$ and its parts

H^{eff}, R (signed), $\Delta E = E_Q - E_P$; refuse $|\Delta E| < 10^{-8}$
$T^{PQ} \leftarrow \frac{1}{2}(R_P R_Q^{\mathsf{T}} - R_Q R_P^{\mathsf{T}})$ ▷ exactly antisymmetric
$D^{\text{PT2}} \leftarrow$ Eq. (51) at T^{PQ} over the handed-over amplitudes
if fold **then**
    $(\gamma, \Gamma, \text{csf}) \leftarrow$ `nac_combination_terms`(T^{PQ}, D^{PT2})
    $F \leftarrow G(W^{PQ})$ with $\Delta E(\gamma, \Gamma)$ and $\Delta E(D^{\text{PT2}} - D^{\text{PT2}\mathsf{T}})$ folded in
    `total` $\leftarrow F/\Delta E +$ csf (per the options)
else
    $\mathbf{h}_{PQ} \leftarrow G(W^{PQ})$
    $\mathbf{d}^{\text{CAS}} \leftarrow$ `df_nac`(T^{PQ}); $\mathbf{d}^{\text{PT2}} \leftarrow$ `df_nac`$(0, D^{\text{PT2}})$
    `total` $\leftarrow \mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\text{CAS}} + \mathbf{d}^{\text{PT2}}$ (per the options)

Conventions. Bra P, ket Q: $\mathbf{d}_{PQ} = -\mathbf{d}_{QP}$, $\mathbf{h}_{PQ} = \mathbf{h}_{QP}$ (PySCF's `kernel(state=(I,J))` is $\langle I|\partial J\rangle$). $\mathbf{d}_{PQ}$ changes sign with the CI phases as R carries them, by $s_{m(P)}s_{m(Q)}$, $m(P)$ the index of R_P's largest component; no phase is fixed by the call.

10 Conical intersections

`qd_*_intersection` returns $\mathbf{g}_P = G(u_P u_P^{\mathsf{T}})$, $\mathbf{g}_Q = G(u_Q u_Q^{\mathsf{T}})$ and $\mathbf{h}_{PQ} = G(W^{PQ})$ from one set of H^{eff} eigenvectors, by default through one several-W call (one shared block Z-vector). It does not refuse a degenerate pair: G is linear in W and nothing divides by $E_Q - E_P$. At degeneracy u_P, u_Q are any orthonormal pair of the two-state space and the three vectors turn with them, but the branching plane $\text{span}\{\mathbf{g}_Q - \mathbf{g}_P, \mathbf{h}_{PQ}\}$ and $\frac{1}{2}(\mathbf{g}_P + \mathbf{g}_Q)$ do not, which is all `geometry::optimize_meci` uses.

Gap sensitivities. An element $a_W = \text{tr}(WH^{\text{eff}})$ is not stationary in the SA-CASSCF parameters, so a solution with residual $\mathcal{G}$ moves it by $\delta a_W = -z^W \cdot \mathcal{G}$ to first order, z^W the W-form's Z-vector. In the pair's 2×2 block $[[a, b], [b, d]]$ the gap is $\sqrt{(a-d)^2 + 4b^2}$, and

$$|\delta \,\text{gap}| \leq s_o \|g\| + s_c \sqrt{n}\, \rho_{\text{CI}}, \qquad s_o = \sqrt{\|z_o^Q - z_o^P\|^2 + 4\|z_o^{PQ}\|^2}, \quad s_c = \sqrt{\|\hat{z}_c^Q - \hat{z}_c^P\|^2 + 4\|\hat{z}_c^{PQ}\|^2}, \tag{58}$$

$\hat{z}_{c,r} = 2w_r z_{c,r}$, $\|g\|$ the SA orbital gradient norm and ρ_{CI} the largest Davidson residual. Θ needs no term (Rayleigh–Ritz) and z^c none (a direct diagonalisation). Both s are invariant to turning u_P, u_Q within their pair.

Inside a degenerate CASSCF pair (#902). A QD pair can be healthy while the CASSCF pair under it is not: at ethylene's C_{3v} MECI the QD gap is 6.6×10^{-10} but the CASSCF ^{1}E pair is $7.9 \times 10^{-9}\, E_h$ apart, below `mcscf::kRootGapFloor` (10^{-8}), where the root-pair multipliers are refused. Write $E_K(R) = \tilde{E}_K(R, \theta(R))$, θ the references' rotation within the pair. $\tilde{E}_K$ is smooth, but at the reference seam θ turns by half the polar angle in the reference branching plane, so $dE_K/dR = \partial_R \tilde{E}_K + \tau_K \partial_R \theta$ with $\partial_R \theta \sim 1/\rho$. SC's g_K diverge; PC's are bounded, but their limit depends on the direction of approach. Only PC's $\tau_P + \tau_Q$ vanishes (measured at $\leq 5 \times 10^{-12}$), so only $\bar{\mathbf{g}} = \frac{1}{2}(\mathbf{g}_P + \mathbf{g}_Q)$ has a limit, and $\bar{\mathbf{g}}$ is all a MECI walk's convergence test reads. So when the closest CASSCF pair is under the floor and carries more than half of u_P's and u_Q's weight,

`qd_pc_intersection` makes one W-form call at $W = \frac{1}{2}(u_P u_P^\mathsf{T} + u_Q u_Q^\mathsf{T})$ with that pair's ζ set to zero and its τ gated at 10^{-11} (`kLimitTauGate`), and returns `g_mean` and `reference_pair_tau` with $\mathbf{g}_P$, $\mathbf{g}_Q$, $\mathbf{h}_{PQ}$ empty and s_o, s_c NaN. `optimize_meci` carries the branching plane from its last evaluation, re-orthonormalised against the rigid motions, and drops the gap from its trust ratio at such an evaluation. SC, and a degenerate pair that is not the QD pair's, keep the refusal. The #847 walk that stopped at evaluation 10 now converges there ($\tau = 5.6 \times 10^{-18}$, QD gap 4.8×10^{-11}), 3.1×10^{-5} Å from exact C_{3v}.

11 Numerical hazards and refusals

- **SC's inactive non-invariance.** SC is not invariant to rotations among inactive orbitals, so z^c divides by orbital-energy gaps (Eq. (24)); a degenerate pair with a finite source is refused. PC's source vanishes with the gap, and its z^c keeps a finite limit there. Park finds SC's gradients unstable for this reason and recommends PC [8].

- **SC's small-norm pole.** A perturber whose own norm is tiny (around 10^{-8} on some symmetric cases) has a Rayleigh quotient H/N that round-off moves, and $c_t = -1/(\Delta + H/N)$ amplifies it by $\sim 1/N$. The energy cannot see it (cN is small); the bars and the PT2 density can. PC divides by no norm and is unaffected. The one-function classes no longer have it: their η (Eq. (5)) reads no N_t.

- **SC's guard.** A perturber under $|N| = 10^{-14}$ is dropped, so SC's energy kinks where a norm passes through zero and is not smooth at a symmetric geometry. The gradient is the one-sided limit; the Hessian refuses the pairs on which it is no bilinear form (§7.5). For the one-function classes, where the guard dropped a perturber whose QD couplings are linear in its amplitude, it is gone (#949).

- **PC's metric cut.** A real mode crossing the cut changes the rank, and the energy by a step of that mode's size (under $10^{-9}\,E_h$ where measured); a non-positive $\lambda + \Delta$ is refused. Near either, the energy is not smooth, and the derivatives inherit the solve's conditioning. First derivatives tolerate kept modes just above the cut; second derivatives do not, hence the floor (§4.3's "conditioning floor") and the Hessian's refusals of a thin kept spectrum and a nearby Schur lift.

- **Inactive near-degeneracies.** z^c and κ_{oo} divide by inactive orbital-energy gaps. In the gradient that is SC's instability and PC's finite limit; in the Hessian κ_{oo} enters quadratically. SC refuses a coordinate whose avoided-crossing half-width is under 10^{-3} bohr. PC's $1/\text{gap}^2$ terms cancel, at a cost in digits, down to the $10^{-6}\,E_h$ degenerate-pair rule.

- **Gaps.** QD states closer than $10^{-8}\,E_h$ are refused by the state form, the coupling and the NAC (not by the W forms or the intersection); CASSCF roots closer than that which ζ, t or T^{PQ} couple are refused, except PC's intersection inside the QD pair's own CASSCF pair, which returns the mean gradient alone (§10).

- **Reproducibility.** The einsums and the DF derivative kernels sum with atomics, so two runs differ at the 10^{-12} relative level (more where a solve is ill-conditioned); `rdms` and its adjoint are bitwise reproducible.

- **Convergence.** Every derivative assumes a converged SA-CASSCF; the gate measures how far the Lagrangian is from stationary, and the gap sensitivities bound the effect on a QD gap.

12 Verification

Each acceptance bar is fixed by analysis and checked once: 10^{-10} for two paths to the same number, 10^{-7} relative for a Richardson ladder of central differences (successive changes falling by a factor in $(3.2, 4.8)$ over three halvings), and twice the committed error bar for a numerical reference.

quantity	checked against
SC energies	PySCF DF-NEVPT2, every class, $10^{-7}\,E_h$
PC energies	block2 `WickICNEVPT2`, every class and dropped count
SC-, PC-QD H^{eff}	a brute-force reference: every element's per-class coupling, 10^{-10}
dm3, f3ca, f3ac	PySCF, 10^{-12} of max; tiled vs one tile, 10^{-13}
RDM adjoints	polarisation identities (the build is quadratic in c, linear in $(tu\|vw)$)
class reverse passes	Eq. (16), homogeneity and Euler's degree 4, 10^{-12}; ladders along B, h_v, ε, h, RDMs
$(\bar{h}, \Lambda, y, A)$	ladders along h, B, each rotation block and c, every input rebuilt
z^c	stationarity of $E_{\text{corr}} + \Phi$ along core–core and virtual–virtual rotations
Lagrangians	stationarity ladders (orbitals, CI off every root, rotations among roots) at equal, unequal and mixed weights; they pin $1/(2w_r)$
gradients	numerical gradients, 2× bar; translation and rotation sums
W form	$G(W_1 + W_2) = G(W_1) + G(W_2)$, 10^{-10}
$\mathbf{h}_{PQ}$, $\mathbf{d}^{\text{NEV}} + \mathbf{d}^{\text{CAS}}$	numerical references, 2× bar; $\mathbf{h}_{QP} = \mathbf{h}_{PQ}$, $\mathbf{d}_{QP} = -\mathbf{d}_{PQ}$
$\mathbf{d}_{PQ}$, $\mathbf{d}^{\text{PT2}}$, D^{PT2}	a brute-force reference, 2× bar; 10^{-10} of max $\|D\|$
fold	the three-solve path, 10^{-9} (two independently converged solves)
intersection	each W form at the same u's; the gap bound on a stopped SA-CASSCF; inside a degenerate CASSCF pair, $\bar{\mathbf{g}}$ against directional differences of $\bar{E}$ at the exact-C_{3v} point
one-function η	$\|X - \eta Y\|/\|X\| \le 6 \times 10^{-15}$ on a state's RDMs; the SC-QD checks that missed at ethylene's Py
t_x, `builders_tangent`	ladders along each input; f against `zc_adjoint`'s Φ
HVPs	ladders of the adjoint along t_B; Euler zeros of Eq. (37); PC at $f = 10^{-10}$ on PSB3; the Schur-lift report against a direct probe of the kept σ
SC, PC Hessians	semi-numerical Hessians of #803's fixtures (two for SC, four for PC), every element within 2× bar (worst 0.085–0.118 bars); each part above 10 bars; CP residual $\le 10^{-8}$, asymmetry $\le 10^{-9}$, translation sum $\le 10^{-8}$
Hessian refusals	each reached by a real input: `nh3_cs_sto3g`'s A″×A″ block and `nevpt2.toml`'s water (SC); a thin kept class, HCN CAS(6,6)'s lift, C_{3v} ammonia's degenerate pair (PC, #943)

A converged residual says a solve answered the equations it was given, not that they were the right ones, so each piece also has a check that reads no residual: forward/reverse duality ($\langle u, y\rangle = \dot{E}(u)$, $\langle \kappa, A\rangle = \dot{E}(\kappa)$, and along a seeded integral perturbation $\langle(\bar{h}, \Lambda), (\dot{h}, \dot{B})\rangle = \dot{E}$); the gate; $E_{\text{corr}} \to 0$, where every multiplier is zero and the gradient is the DF-CASSCF one; stationarity ladders of $\mathcal{L}$; and the translational and rotational sums.

The oracle. No program the tree can import has a NEVPT2 gradient, so the end-to-end oracle is the numerical gradient of re-converged energies, at the same settings and the same auxiliary basis (an approximation is tested against the same approximation). PySCF's DF-NEVPT2 gives SC, and block2's or this tree's energy gives PC. Each displaced CASSCF passes a continuity guard before it is differenced: its energy against the first-order prediction, its active subspace and CI vector against the reference's, and convergence. A displaced optimisation that lands on another stationary point gives a number from another reference. Central differences plus one Richardson step are good to about $10^{-6}\,E_h/a_0$, and that is the gate a consumer holds an analytic gradient to, not the tree's

usual 10^{-7}, because the oracle is itself a difference. Near-degenerate inactive orbitals (SC's hazard, §11) are where it fails first: the analytic gradient is right there, but the energy curves too sharply for the differences.

13 Map to the code and to the papers

here	code
§4.1	`nevpt2::mo_blocks`, `h2e`, `slab`, and `*_adjoint`
§4.2	`nevpt2::rdms`, `rdms_adjoint` (one vector; bra/ket)
Eqs. (4), (6)	`class_energies`, `pc_class_energies`
Eqs. (7)–(8)	`sc_effective_hamiltonian`, `pc_effective_hamiltonian`
§5	`class_energies_adjoint`, `pc_class_energies_adjoint`, `*_effective_hamiltonian_adjoint`
Eqs. (30), (31), (33)	`response::zvector`; `mcscf::df_gradient`'s multipliers and extra adjoint
§6	`sc_energy`, `sc_adjoint`, `pc_*`, `zc_multipliers`, `zc_adjoint`, `lagrangian_gradient`, `sc_gradient`, `pc_gradient`
§8	`qd_*_energy`, `qd_*_adjoint`, `qd_sc_lagrangian`, `qd_*_gradient` (state, W, several W)
§9	`qd_*_coupling`, `*_pt2_transition_density`, `qd_*_nac`
§10	`qd_*_intersection`
Eq. (5)	`single_function_etas`, `single_function_eta_bars`
§4.3, the floor	`EnergyInputs::pc_floor`, `PcClassEnergies::cut`, `kept`
§4.5	`Nevpt2Step::run` (`pipeline/nevpt2`); `sc_couplings`, `pc_couplings`, `pc_kets`, `solve_class`
Eq. (36), Eq. (35)	`nevpt2::builders_tangent`, `hessian::df_nevpt2_tangents`
§7.3	`class_energies_hvp`, `pc_class_energies_hvp`, `sc_hessian_product`, `pc_hessian_product`
§7.4	`hessian::df_sc_nevpt2`, `df_pc_nevpt2`, `df_adjoint_xy`, `df_casscf`, `df_casscf_z`
§7.6	`pipeline/hessian`'s numerical method, ladder and bilinearity check; `tools/nevpt2_hessian`

here	Park
Eq. (15), Eq. (16)	[8], Eqs. 36–38, 40
Eq. (24)	[8], Eqs. 56–57; [10], Eqs. 25–27
Eq. (29)	[8], Eq. 23
Eq. (9), Eq. (43)	[10], Eqs. 16–20 ($R_{PPMN} = W_{MN}$)
the QD reverse passes	[10], Eqs. 21–23 (λ, implicit in the bars), 31–43
Eq. (44)	[10], Eq. 24, with ζ and Θ explicit
the Z-vector right-hand sides	[10], Eqs. 28–29
§9	[10], Eqs. 45–54

Where this departs from Park.

- Park's QD gradient is PC only, with SC called straightforward [10]; both are implemented here, with SC's hazards stated.
- Park reports gradients and optimised geometries, not Hessians. The single-state Hessian of §7 has no counterpart in his papers; its only reference is the semi-numerical Hessian over the analytic gradient.

- SC's one-function classes keep every perturber at η (Eq. (5)) instead of PySCF's guard, so this SC differs from PySCF's and Park's where a one-function perturber's norm passes through zero (by at most $6.5 \times 10^{-13}\,E_h$ where measured).
- Park contracts 4-RDM derivatives through multipass intermediates ([10], Eqs. 55–58). Here the f3ca/f3ac build is differentiated directly, tile by tile, so neither the 4-RDM nor its derivative exists. Kollmar et al.'s factorisation of the Dyall matrix elements [12] would remove f3 altogether; it is not used.
- ε is a function of γ (or γ^{SA}) inside the functional, the multipliers among the roots (ζ, Θ) are explicit, and the reference block is the diagonal form $E_M \delta_{MN}$, equal to Park's on the constraint surface.
- Printed misprints affect a reader of [10]: Eq. 5 drops the two-electron integral of the active term; Eq. 16 omits $\langle M|H|N\rangle$; Eq. 29's $z^c_{xy}\langle I|E_{xy}|N\rangle$ is zero as printed and should be $[\mathbf{g}(z^c)]_{xy}$ ($\mathcal{W}$ here); Eq. 41 has $\Gamma^{(3)}_{a'a,b'b,ec}$ where Eq. 39 has $\Gamma^{(3)}_{a'd,b'b,ec}$; Eq. 56 sums over its free indices.

References

[1] C. Angeli, R. Cimiraglia, S. Evangelisti, T. Leininger and J.-P. Malrieu, Introduction of n-electron valence states for multireference perturbation theory, J. Chem. Phys. **114**, 10252 (2001), doi:10.1063/1.1361246.

[2] C. Angeli, R. Cimiraglia and J.-P. Malrieu, N-electron valence state perturbation theory: a fast implementation of the strongly contracted variant, Chem. Phys. Lett. **350**, 297 (2001), doi:10.1016/S0009-2614(01)01303-3.

[3] C. Angeli, R. Cimiraglia and J.-P. Malrieu, n-electron valence state perturbation theory: a spinless formulation and an efficient implementation of the strongly contracted and of the partially contracted variants, J. Chem. Phys. **117**, 9138 (2002), doi:10.1063/1.1515317.

[4] C. Angeli, S. Borini, M. Cestari and R. Cimiraglia, A quasidegenerate formulation of the second order n-electron valence state perturbation theory approach, J. Chem. Phys. **121**, 4043 (2004), doi:10.1063/1.1778711.

[5] K. G. Dyall, The choice of a zeroth-order Hamiltonian for second-order perturbation theory with a complete active space self-consistent-field reference function, J. Chem. Phys. **102**, 4909 (1995), doi:10.1063/1.469539.

[6] A. Zaitsevskii and J.-P. Malrieu, Multi-partitioning quasidegenerate perturbation theory. A new approach to multireference Møller–Plesset perturbation theory, Chem. Phys. Lett. **233**, 597 (1995), doi:10.1016/0009-2614(94)01503-N.

[7] A. A. Granovsky, Extended multi-configuration quasi-degenerate perturbation theory: the new approach to multi-state multi-reference perturbation theory, J. Chem. Phys. **134**, 214113 (2011), doi:10.1063/1.3596699.

[8] J. W. Park, Analytical gradient theory for strongly contracted (SC) and partially contracted (PC) N-electron valence state perturbation theory (NEVPT2), J. Chem. Theory Comput. **15**, 5417 (2019), doi:10.1021/acs.jctc.9b00762.

[9] J. W. Park, the preprint of Ref. [8], arXiv:1907.10180, doi:10.48550/arXiv.1907.10180.

[10] J. W. Park, Analytical gradient theory for quasidegenerate N-electron valence state perturbation theory (QD-NEVPT2), J. Chem. Theory Comput. **16**, 326 (2020), doi:10.1021/acs.jctc.9b00919.

[11] J. W. Park and T. Shiozaki, Analytical derivative coupling for multistate CASPT2 theory, J. Chem. Theory Comput. **13**, 2561 (2017), doi:10.1021/acs.jctc.7b00018.

[12] C. Kollmar, K. Sivalingam, Y. Guo and F. Neese, An efficient implementation of the NEVPT2 and CASPT2 methods avoiding higher-order density matrices, J. Chem. Phys. **155**, 234104 (2021), doi:10.1063/5.0072129.

[13] N. C. Handy and H. F. Schaefer III, On the evaluation of analytic energy derivatives for correlated wave functions, J. Chem. Phys. **81**, 5031 (1984), doi:10.1063/1.447489.

[14] P. J. Knowles and N. C. Handy, A new determinant-based full configuration interaction method, Chem. Phys. Lett. **111**, 315 (1984), doi:10.1016/0009-2614(84)85513-X.

[15] O. Vahtras, J. Almlöf and M. W. Feyereisen, Integral approximations for LCAO-SCF calculations, Chem. Phys. Lett. **213**, 514 (1993), doi:10.1016/0009-2614(93)89151-7.

[16] Q. Sun *et al.*, Recent developments in the PySCF program package, J. Chem. Phys. **153**, 024109 (2020), doi:10.1063/5.0006074.

[17] H. Zhai *et al.*, Block2: a comprehensive open source framework to develop and apply state-of-the-art DMRG algorithms in electronic structure and beyond, J. Chem. Phys. **159**, 234801 (2023), doi:10.1063/5.0180424.

[18] R. D. Guerrero, Analytic gradients and nonadiabatic couplings for device-resident DMRG-QD-NEVPT2 through conical intersections on a consumer GPU, arXiv:2609.09990 (2026), doi:10.48550/arXiv.2609.09990.